

CORE MATHEMATICS GRADE 10: REAL NUMBERS

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.1 (8 Lessons)

ATTRIBUTION AND LICENSING

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Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Core Mathematics
Strand	Strand 1.0: Numbers and Algebra
Sub-Strand	Sub-Strand 1.1: Real Numbers
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Types of real numbers: natural numbers, integers, rational numbers, irrational numbers and their properties – Number line representation of real numbers including placement of surds and irrational numbers – Operations on real numbers: addition, subtraction, multiplication and division – Order of operations (BODMAS/BIDMAS) applied to real numbers in multi-step computations – Absolute value of real numbers and its applications in real-life contexts – Reciprocals of real numbers and their use in mathematical computations – Conversion between fractions, decimals and percentages within the real number system – Approximation and estimation: rounding off, significant figures and decimal places – Application of real numbers to real-life situations (e.g., temperatures in Nairobi, distances in the Rift Valley, market prices in Gikomba)
Learning Outcomes	By the end of the sub-strand, the learner should be able to: - classify and describe types of real numbers and their properties, - represent real numbers on a number line, - perform operations on real numbers accurately, - determine the absolute value of real numbers and apply it in mathematical computations,

	e) determine the reciprocal of real numbers and accurately apply the reciprocal of real numbers in mathematical computations, f) apply approximation and estimation skills (rounding off, significant figures, decimal places) in real-life contexts, g) apply real numbers in solving real-life mathematical problems, h) appreciate the use of real numbers in day-to-day life situations.
Core Competencies	– Communication and Collaboration: the learner discusses and justifies the classification of numbers encountered in everyday Kenyan contexts — such as market prices at Gikomba market or temperatures recorded at Wilson Airport — working in pairs and groups to share reasoning about the real number system. – Critical Thinking and Problem Solving: the learner evaluates, compares and operates on real numbers to solve multi-step problems drawn from real-life situations such as calculating distances between Nairobi and Mombasa or working out fair-price divisions of maize harvests in the Rift Valley. – Self-Efficacy: the learner develops confidence in computing with real numbers — including reciprocals, absolute values and approximations — by tackling increasingly complex tasks and reflecting on their own accuracy and strategies. – Digital Literacy: the learner uses calculators and, where available, digital devices to verify computations involving real numbers, significant figures and reciprocals, and to explore number patterns.
Core Values	– Integrity: the learner records and reports computations — such as approximated measurements or market calculations — accurately and honestly, without manipulating results. – Responsibility: the learner takes ownership of their mathematical reasoning, carefully checking operations on real numbers and acknowledging errors during peer-review activities. – Unity: the learner collaborates respectfully with peers from diverse backgrounds during group classification and computation tasks, valuing each member's contribution to problem solving. – Love: the learner shows care for fellow learners by offering support and explanation when peers struggle with concepts such as identifying irrational numbers or applying BODMAS to real numbers.
Science & Engineering Practices	– Asking Questions and Defining Problems: the learner poses questions about why certain numbers (e.g., $\sqrt{2}$, π) cannot be expressed as exact fractions, driving inquiry into the structure of the real number system. – Using Mathematics and Computational Thinking: the learner applies order of operations, absolute value rules and reciprocal relationships to solve multi-step real-world problems, mirroring the systematic reasoning used in scientific and engineering contexts. – Analysing and Interpreting Data: the learner interprets numerical data from Kenyan real-life contexts (e.g., rainfall figures from the Kenya Meteorological Department, population data, commodity prices) by classifying, approximating and operating on the values using properties of real numbers. – Constructing Explanations: the learner constructs clear written and verbal explanations of why a number belongs to a particular subset of the real number system, using evidence from the number line and computation results.
Pertinent & Contemporary Issues (PCIs)	– Financial Literacy: the learner applies operations on real numbers — including reciprocals, percentages and approximations — to real-life financial contexts such as calculating unit prices of unga (maize flour) or sukuma wiki at a Nairobi market, making change, and estimating household budgets, thereby building practical money-management skills. – Education for Sustainable Development (Environmental Education): the learner uses real-number computations to engage with environmental data, for example approximating distances for tree-planting initiatives in Karura Forest or calculating ratios of land use in the Rift Valley, linking mathematics to sustainability. – Life Skills (Decision Making): the learner applies estimation and approximation skills to make informed everyday decisions — such as determining whether a quoted price at the duka is a fair approximation or whether a measured quantity of githeri ingredients is within an acceptable range. – Social Cohesion: the learner works collaboratively in diverse groups to classify numbers and solve real-life problems, fostering

	mutual respect and appreciation across different community backgrounds.
Career Connections	– Engineering and Architecture: engineers working on infrastructure projects such as the Nairobi Expressway or LAPSET Corridor apply real number operations, significant figures and approximations in precise calculations of measurements, loads and material quantities. – Data Science and Actuarial Science: data analysts and actuaries at institutions like the Kenya Revenue Authority (KRA) or insurance companies use properties of real numbers, rational/irrational distinctions and precision in decimal representation to model financial and population data. – Accounting and Finance: accountants and bank officers working in institutions such as Equity Bank or Kenya Commercial Bank apply reciprocals, percentages, rounding off and operations on real numbers daily in ledger computation, interest calculations and currency exchange. – Agriculture and Agribusiness: agronomists and farmers in the Rift Valley or around Kisumu apply real-number arithmetic to calculate fertiliser ratios, land areas, crop yield estimates and fair market pricing for produce like maize, tea or beans. – Medicine and Pharmacy: medical practitioners at Kenyatta National Hospital and pharmacists apply exact and approximate real-number computations when calculating drug dosages, body-mass indices and lab result thresholds.
Focus for Lessons	This unit anchors the Grade 10 Core Mathematics Number and Algebra strand by building a rigorous yet accessible understanding of the real number system — encompassing natural numbers, integers, rationals and irrationals — and the operations, approximations and applications that flow from it. Learners engage with Kenyan real-life contexts (market prices, land measurements, temperature data and distance problems) to make the abstract structure of real numbers tangible and meaningful. The instructional approach is inquiry-based and collaborative: learners predict, investigate, explain and model their growing understanding across eight progressive lessons, each connecting back to the anchoring phenomenon of encountering numbers in everyday Kenyan life that cannot always be expressed exactly as simple fractions.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why can we never write some everyday numbers down completely — and how do mathematicians and Kenyan professionals work accurately with numbers that go on forever? KICD KEY INQUIRY QUESTIONS: 1. What are real numbers? 2. How do we use real numbers in day-to-day situations?
Anchoring Phenomenon	ANCHORING PHENOMENON — "The Ugali Pot Problem": A mama mboga in Gikomba Market, Nairobi, uses a cylindrical aluminium sufuria to cook ugali for her family every evening. She knows the sufuria has a diameter of exactly 28 cm. When her daughter — a Grade 10 learner — tries to calculate the exact circumference of the pot to cut a piece of string that will fit perfectly around the rim, she multiplies 28 by π (pi). Her calculator shows 87.96459430... and the digits never end or repeat. She cuts the string to 88 cm (rounded to the nearest centimetre), but there is always a tiny gap or overlap. What is puzzling: The diameter is a perfectly "nice" whole number — 28 cm — yet the circumference cannot be written as an exact fraction or a terminating decimal. No matter how many decimal places the calculator shows, the number goes on forever without repeating. Meanwhile, other everyday measurements — the price of a kilo of unga (Ksh 60.50), the distance from Nairobi to Nakuru (160 km), the temperature in Kisumu (34.5°C) — seem to be expressible as ordinary fractions. Why can some numbers be written exactly while others cannot? What kinds of numbers actually exist, and how do we work with them accurately in real life when we can never write them down completely?

	This phenomenon propels learners to classify, represent and operate on the full system of real numbers, understand the role of approximation, and appreciate why mathematicians draw a fundamental distinction between rational and irrational numbers.
Storyline Thread	Lesson 1 — "What Kind of Number Is That?" (Predict & Classify) Learners encounter the Ugali Pot anchoring phenomenon. They predict: is π a fraction? Is it a whole number? Working in pairs, they sort a set of 20 Kenyan number cards (prices, distances, temperatures, $\sqrt{2}$, π, -7, 0, $2/3$, $0.333\dots$) into groups using their own invented categories. Groups share their sorting rules. Teacher introduces the formal classification: natural numbers $\subset$ integers $\subset$ rationals $\subset$ real numbers, and identifies irrationals. Learners revisit their cards and reclassify. DQB is opened: learners post their initial questions. Lesson 2 — "Placing Numbers in Their Right Home" (Number Line & Representation) Learners construct a large classroom number line (using string and pegs) spanning -5 to 5. They must physically place cards showing $-\sqrt{3}$, -1.5, 0, $2/3$, $\sqrt{2}$, $\pi - 3$, $2.333\dots$, and 3 onto the line, justifying each placement. The supporting phenomenon of the Kericho temperature record is introduced: learners plot the week's temperatures and identify which are rational vs. irrational. Learners update the DQB: "Can we always find the exact spot for an irrational number?" Lesson 3 — "Operating on Real Numbers" (Operations & BODMAS) Learners apply the four operations to real numbers using Kenyan market contexts (e.g., calculating total cost of ugali flour, sukuma wiki and tomatoes with mixed-number quantities). Focus on BODMAS with multi-step expressions. Pairs solve sets of increasingly complex expressions, then peer-mark using worked solutions. Common errors are surfaced and discussed. Connection back to phenomenon: "When we approximate π to compute the circumference, which operation rules still apply?" Lesson 4 — "How Far From Zero?" (Absolute Value) The Kericho temperature supporting phenomenon is revisited: "Which day had the biggest swing away from zero degrees?" Learners discover absolute value as "distance from zero on the number line." They work through formal definition x, compute absolute values of real numbers, solve simple absolute value expressions in context (e.g., $\text{temperature change}$, $\text{profit/loss in shillings}$), and represent results on the number line. DQB updated: "Does absolute value change whether a number is rational or irrational?" Lesson 5 — "Flipping Numbers: Reciprocals" (Reciprocals & Their Properties) The boda-boda fuel-split supporting phenomenon is introduced. Learners explore: if one rider pays $850/3$ shillings, what is the reciprocal of $850/3$? What does multiplying a number by its reciprocal always give? Learners derive the rule, practise finding reciprocals of integers, fractions, decimals and simple surds, and apply reciprocals in context (unit rates, speed calculations for the Iten marathon runner phenomenon). Learners test: does every real number have a reciprocal? (Discovery: zero does not.) Lesson 6 — "Close Enough? Approximation and Significant Figures" (Rounding, S.F., D.P.) Returning to the sufuria phenomenon: the daughter rounded $87.9645\dots$ to 88 cm. Was that wise? Learners investigate rounding to decimal places, significant figures, and when to round up vs. down in real-life contexts (drug dosages at KNH, land measurements in the Rift Valley, fuel costs). They compare the effect of rounding at different stages of a multi-step calculation (e.g., computing the total cost of fencing a maize shamba using π and $\sqrt{2}$). DQB updated: "Does rounding ever make an irrational number rational?" Lesson 7 — "Real Numbers in Real Kenya" (Application & Problem Solving)

	Learners work in groups on a structured problem-solving task that integrates all sub-strand content: a scenario in which a community water project near Lake Victoria needs to calculate pipe lengths (involving surds), water volumes (operations on decimals), cost estimates (approximation to significant figures), and equal cost-sharing (reciprocals and rational arithmetic). Each group produces a written solution with justifications. Groups present key findings to the class. Teacher facilitates sensemaking connections back to the DQB and anchoring phenomenon. Lesson 8 — "Building Our Real Number Model" (Model Building, DQB Closure & Assessment) Learners individually construct a cumulative visual model of the real number system: a nested Venn diagram annotated with Kenyan examples in each region, a number line showing at least one example from each sub-category, and a worked example showing how approximation, absolute value and reciprocals each appear in a single real-life Kenyan problem of their own devising. DQB is reviewed: which questions have been answered? Which remain open (pointing forward to indices and logarithms)? Lesson concludes with a class written assessment task.
Supporting Phenomena	– The Maize Field Diagonal: A farmer in the Rift Valley has a perfectly square maize shamba with sides of exactly 10 metres. She wants to string a fence diagonally across the field. Using Pythagoras, the diagonal is $\sqrt{200} = 10\sqrt{2} \approx 14.142\dots$ metres — again a non-terminating, non-repeating decimal. Yet the side length is a "nice" number. Learners explore why $\sqrt{2}$ is irrational and where it sits on the number line relative to 1.4, 1.41, 1.414, etc. – The Kericho Tea Estate Temperature Record: The Kenya Meteorological Department records daily minimum temperatures in Kericho. On a particular week, the recorded temperatures (in °C) are: -2.5, 0, 3, -1, 4.75, $2\frac{1}{3}$, $\sqrt{3}$. Learners must classify each temperature as natural, integer, rational or irrational, place them on a number line, and compute the absolute differences between consecutive days to find the sharpest overnight temperature change. – The Boda-Boda Fuel Split: Three boda-boda riders in Kisumu pool money to buy fuel. The total cost is Ksh 850. They must split the bill equally. Dividing $850 \div 3 = 283.333\dots$ (a repeating decimal, hence rational but non-terminating). Learners explore why this is rational (it can be written as $850/3$), contrast it with irrational numbers, and practise rounding to the nearest shilling for a fair, practical split. – The Nairobi Marathon Pace: A Kenyan marathon runner from Iten trains for the Nairobi Standard Chartered Marathon. Her target race pace is 1 km every 3 minutes and 45 seconds. Learners convert mixed units, apply reciprocals (speed = $1 \div \text{time-per-km}$) to find km per minute, and use significant figures to compare her pace to competitors, connecting real-number operations to athletics and career contexts.

LESSON 1 (80 minutes): What Kind of Number Is That? (Predict & Classify)

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	This opening lesson launches the anchoring phenomenon — the Ugali Pot Problem — by surfacing learners' prior knowledge of different types of numbers and creating productive cognitive dissonance around the number π . Learners predict, sort, and begin to classify everyday numbers, discovering that some numbers cannot be written exactly, which propels the entire unit's inquiry.
Knowledge	Learners will know that: (1) numbers can be classified into different types (whole numbers, fractions, decimals, and numbers like π that do not terminate or repeat); (2) the real number system contains both rational and irrational numbers; (3) everyday Kenyan measurements and quantities can be categorised within the real number system.
Skills	Learners will be able to: (1) sort a given set of numbers into 'can be written exactly' and 'cannot be written exactly' categories using reasoning; (2) represent numbers on a number line; (3) articulate in writing and speech why π presents a challenge that ordinary fractions do not; (4) pose investigative questions for the Driving Question Board.
Attitudes	Learners will appreciate that: (1) mathematics is a tool for making sense of real Kenyan professional and everyday life situations; (2) asking 'what kind of number is that?' is a powerful mathematical habit of mind; (3) the inability to write certain numbers exactly is not a failure of mathematics but a profound and fascinating truth about the number system.
Key Inquiry Question	1. What are real numbers? 2. How do we use real numbers in day-to-day situations?
Purpose in Storyline	Lesson 1 is the phenomenon-launch lesson. It drops learners into the Ugali Pot Problem, elicits their existing number intuitions through a predict-and-sort task, surfaces the central puzzle (why is $28 \times \pi$ unwritable yet 28 is perfectly 'nice'?), and opens the Driving Question Board that will anchor all subsequent lessons in the unit. No formal definitions are given yet — the lesson is designed to create curiosity, not closure.
Safety Notes	No hazardous materials or activities in this lesson. If physical sufuria/pots are brought to class, ensure edges are not sharp. Ensure calculators or learner digital devices are handled responsibly and returned after use.

B. LESSON OVERVIEW

Learners are introduced to the anchoring phenomenon — the Ugali Pot Problem from Gikomba Market — through a short story read-aloud and a visual display of the sufuria. They first predict (without any instruction) whether a set of ten carefully chosen everyday Kenyan numbers can or cannot be written down completely and exactly. They then observe the phenomenon directly by computing $28 \times \pi$ on a calculator and comparing it with 28×3 or 28×0.5 , noticing the endless, non-repeating decimal. Pairs share and justify their sorting decisions, generating rich disagreement that surfaces misconceptions. The lesson closes with learners co-constructing the opening Driving Question Board (DQB), posting their most burning questions about numbers that 'go on forever'. No formal vocabulary (rational/irrational) is introduced — that is reserved for Lesson 2 — keeping this lesson firmly in the Predict and initial Observe phases.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners listen to the teacher read	1. Before distributing cards, read	Elicit-Before-Explain (Predict	Observable indicators: (1) Can the

 VIDEO: Compare rational numbers using a number line Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Classify numbers: rational & irrational Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	the Ugali Pot Problem story aloud (Gikomba Market, Nairobi; mama mboga; 28 cm sufuria; daughter's string-cutting dilemma). A large image of a cylindrical aluminium sufuria is displayed at the front. Each learner then receives a 'Number Sort Card' containing 10 numbers drawn from Kenyan everyday contexts: (1) 60.50 — price of 1 kg unga in Ksh; (2) 160 — distance Nairobi to Nakuru in km; (3) 34.5 — temperature in Kisumu in °C; (4) $\frac{1}{3}$ — fraction of a githeri portion; (5) 0.333... — decimal form of $\frac{1}{3}$ on a calculator; (6) $\sqrt{2}$ — diagonal of a $1\text{m} \times 1\text{m}$ shamba plot; (7) π — ratio used in the sufuria problem; (8) $\frac{22}{7}$ — common approximation of π; (9) 87.96459430... — calculator result of $28 \times \pi$; (10) -5 — temperature drop at the top of Mt Kenya in °C. Working individually (5 minutes), learners sort the 10 numbers into two columns on their card: Column A — 'I can write this number down completely and exactly' and Column B — 'I cannot write this number down completely and exactly'. Learners also write one sentence explaining their most puzzling choice.	the Ugali Pot Problem story aloud slowly and dramatically, pausing after '...the digits never end or repeat.' Ask: 'Has anyone ever seen a calculator give an answer that just keeps going?' (Show of hands only — no discussion yet.) 2. Distribute Number Sort Cards face-down. Say: 'Do NOT flip the card until I say go. This is your thinking — no calculators yet, no neighbours. Trust what you already know.' 3. Say 'Go.' Set a 5-minute visible timer. Circulate silently, noting which numbers generate the most hesitation (expected: 0.333..., $\frac{22}{7}$, $\sqrt{2}$). Do NOT correct or affirm during this phase. 4. After 5 minutes, say: 'Circle the ONE number on your card that surprised you most — the one you are least sure about.' Allow 30 seconds. 5. Cold-call exactly 4 learners (select 1 high-confidence, 1 mid, 1 struggling, 1 who circled an unexpected item): 'Amina, which column did you put $\frac{22}{7}$ in, and why?' WAIT 12 seconds after each question before accepting a response. Do not affirm or correct — respond only with 'Interesting — does anyone have a different placement?' 6. Record on the board in two columns every placement mentioned, including contradictions, so the class can see genuine disagreement. Say: 'Hold all your placements lightly — we are about to gather some evidence.'	Phase): By requiring individual written predictions before any instruction or peer discussion, the teacher surfaces genuine prior knowledge and misconceptions. Recording contradictory placements visibly on the board creates cognitive dissonance — a necessary precondition for meaningful learning. This strategy also gives the teacher instant formative data about where misconceptions cluster (typically around repeating decimals and common fractions approximating π).	learner distinguish between a terminating decimal (34.5) and a non-terminating one (0.333...) at the prediction stage? (2) Does the learner place $\frac{22}{7}$ in Column A or B — and can they articulate why? (3) Does the learner treat 87.96459430... and π as the same or different kinds of numbers? Teacher records on a clipboard the names of learners who place $\frac{1}{3}$ and 0.333... in different columns (a key misconception to address in Lesson 3) and learners who confidently place $\sqrt{2}$ in Column B (indicating stronger prior knowledge of surds).
Observe Phase  VIDEO: Worked example: classifying numbers	Learners now use calculators (or learner digital devices) to investigate three specific computations drawn directly from the sufuria phenomenon. Each learner records results in their	1. Before calculators are distributed, say: 'We are now gathering evidence — just like a Kenyan engineer at a construction site would gather measurements before making a decision. Record	Evidence-to-Explanation Bridge (Observe Phase): The structured three-computation sequence moves learners from familiar, terminating results ($28 \times 3 = 84$; $28 \times 0.5 = 14$) to the anomalous result	Observable indicators: (1) Does the learner correctly identify that 28×3 and 28×0.5 terminate while $28 \times \pi$ does not? (2) Does the learner's hypothesis contain any proto-mathematical language

Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Classify numbers Source: Khan Academy (English - US curriculum) Search ARES for similar readings	exercise book under the heading 'What I Observed'. The three computations are: (A) $28 \times 3 = ?$ (B) $28 \times 0.5 = ?$ (C) $28 \times \pi = ?$ For each result, learners answer three questions in writing: (i) How many decimal places does this answer have? (ii) Do the decimal digits form a pattern that repeats? (iii) Could you, in theory, write this number out completely on a piece of paper? After individual recording, learners compare results with a partner and mark any of the 10 Number Sort Card items they now wish to move between columns. The teacher then plays a 90-second visual: the digits of π scrolling on the board (pre-loaded: 3.14159265358979323846...) while learners watch in silence. After the scroll, the teacher asks: 'The diameter of the sufuria is 28 — a whole number. A whole number times π. Why can't the answer be a whole number, or at least a neat fraction?' Learners write a hypothesis (2–3 sentences) in their exercise books.	everything you see, even if it confuses you.' 2. Distribute calculators. Give 4 minutes for the three computations. Circulate and watch for learners who type 22/7 instead of the π button — gently redirect: 'Use the π key, not 22/7. Are they the same? That is actually one of our questions today.' 3. After computations, display the π digit scroll for exactly 90 seconds. Stand at the front in silence. Let the visual speak. 4. After the scroll ends, ask: 'What do you notice?' WAIT 15 seconds — a full, uncomfortable silence. Then cold-call 3 learners: 'Ochieng, what did you notice?' / 'Wanjiru, anything different from what Ochieng said?' / 'Kiprop, can you add to that?' Write their exact words on the board without paraphrasing. 5. Pose the hypothesis question: 'Why can't $28 \times \pi$ be a neat fraction or terminating decimal? Write your best hypothesis — not the answer, your best current guess. 3 sentences. Go.' Allow 3 minutes of silent writing. 6. Say: 'You have just done what mathematicians do — gather evidence and form a hypothesis. We will test this hypothesis across many lessons. Do NOT erase your hypothesis — you will compare it to your thinking at the end of the unit.'	($28 \times \pi = 87.96459430\dots$), making the contrast viscerally concrete. The π digit scroll is a deliberate sensory anchor — learners will remember the visual of the endless digits when the unit later formalises the definition of irrational numbers. The hypothesis-writing step initiates the explanation phase at a personal level before class-wide sensemaking begins.	(e.g., 'it never stops', 'it has no pattern', 'you can't make it a fraction') — evidence of emerging conceptual understanding? (3) Does the learner revise their Number Sort Card placements after the calculator activity, and which items are revised? (Teacher notes revisions as evidence of responsiveness to evidence — a key scientific practice.)
Explain Phase VIDEO: Classifying numbers Source: Khan Academy (English - US curriculum) Search ARES for similar videos	Learners form groups of four. Each group receives a large A3 sheet with a blank two-column table (Column A: 'Can be written exactly' / Column B: 'Cannot be written exactly') and the same 10 numbers from the Number Sort Card, this time as cut-out slips. Groups physically place the slips and must reach consensus. Every	1. Form groups of four deliberately (mix confidence levels). Distribute A3 sheets and cut-out number slips. Say: 'You have 8 minutes. Every number must be placed. You must agree. You must have a one-sentence reason for every placement. If you disagree, you must argue your case with evidence — not just opinion.' Set	Structured Academic Controversy (Explain Phase): By requiring groups to reach consensus on contested placements and then exposing disagreements through the gallery walk, the teacher engineers productive intellectual conflict. This is more powerful than direct instruction because learners must articulate and defend	Observable indicators: (1) Can the group correctly identify that 1/3 and 0.333... represent the same number — and place them together? (2) Do learners articulate that 22/7 is a fraction (therefore 'exact' in one sense) but only an approximation of π? (3) Does the learner's written personal response to 'what property would a number

 EXERCISE: Compare rational numbers using a number line Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	placement requires a spoken justification from at least one group member — the group secretary records the justification in one sentence next to each number. Groups then rotate to view two other groups' tables (gallery walk, 3 minutes) and place a sticky dot next to any placement they disagree with. Whole-class debrief follows: teacher highlights the three most contested placements (expected: $1/3$ vs. $0.333\dots$; $22/7$ vs. π; $\sqrt{2}$). Teacher introduces — for the first time — the question: 'What property would a number need to have so that you could write it down completely?' Learners write a personal response (2 sentences) before the class discusses. Teacher draws a simple two-region diagram on the board (no labels yet — just Region A and Region B) and asks learners to suggest which of the 10 numbers belongs in which region, connecting back explicitly to the sufuria: 'Where does the circumference of mama mboga's sufuria live — Region A or Region B?'	timer. 2. During group work, visit each group. Listen for: (a) confusion between $1/3$ (exact fraction) and $0.333\dots$ (its decimal representation — actually the same number); (b) learners claiming $22/7 = \pi$ (a key misconception). Do NOT resolve these — instead, ask: 'Is $22/7 \times 7$ exactly equal to 22? What does that tell you?' WAIT 10 seconds and move on. 3. Signal gallery walk. Say: 'Walk silently. Read. Place a sticky dot on any placement you disagree with. You have 3 minutes.' 4. Return to seats. Display a tally of sticky dots on the board. Ask: 'The three most contested numbers are [name them]. Let's hear the arguments.' Cold-call 2 groups per contested number — one who placed it in A, one in B. WAIT 12 seconds per cold-call. 5. After debate, do NOT give the answer. Say: 'We do not yet have the vocabulary to settle this argument — but we will. What I can tell you is this: mathematicians did settle this argument, and it took thousands of years. By the end of this unit, you will be able to settle it in seconds.' 6. Draw the two-region diagram. Ask: 'Where does the circumference of the sufuria go?' Cold-call 1 learner. Record their answer. Say: 'We will return to this diagram in every lesson and add more detail.'	reasoning, exposing the logical gaps that formal vocabulary (rational/irrational) will later fill. The two-region diagram on the board becomes the seed of the number-system model learners will develop across the unit.	need to be writable exactly?' contain the ideas of 'stopping' or 'repeating' — proto-understanding of terminating and repeating decimals? (4) During gallery walk, do learners engage critically with other groups' placements or merely observe? Teacher marks names of learners who show proto-understanding of the terminating/repeating distinction for extension follow-up.
Driving Question Board (DQB) Creation  VIDEO: Proof: there's an irrational number between any two	Each learner receives two sticky notes (different colours: yellow = a question I have about numbers; blue = a connection I see to real Kenyan life). Learners individually write: on yellow — their single most burning question sparked by the Ugali Pot Problem and today's	1. Distribute sticky notes. Say: 'Yellow is for YOUR questions — not questions you think the teacher wants. Real questions. Things that are actually confusing or curious to you after today. Blue is for connections — where in Kenya, in real jobs, in real life, does this	Driving Question Board Protocol (DQB Phase): The DQB externalises learner curiosity and creates a public, persistent record of the class's collective inquiry. By distinguishing learner-generated questions (yellow) from real-life connections (blue), the teacher	Observable indicators: (1) Are learner questions on yellow notes specific and mathematically substantive (e.g., 'Can every fraction be turned into a decimal that eventually repeats?' 'Is there a number between π and $22/7$?') or vague? Quality of questions is a

rational numbers Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Rational number word problems Source: Khan Academy (English - US curriculum) Search ARES for similar readings	activities; on blue — one Kenyan professional, job, or everyday situation where this kind of number puzzle might matter. Learners post their sticky notes on the class Driving Question Board (a large section of wall or a dedicated poster). The teacher reads aloud 6–8 yellow questions, grouping similar ones in real time. The class votes (show of hands) on the top 3 questions they most want answered by the end of the unit. These 3 questions are written in large print at the top of the DQB and will remain visible for the entire unit. The teacher then posts the unit's official Driving Question beneath the learner questions: 'Why can we never write some everyday numbers down completely — and how do mathematicians and Kenyan professionals work accurately with numbers that go on forever?'	number puzzle appear?' Allow 4 minutes of silent writing. 2. Circulate and read over shoulders. If a learner writes a very shallow question (e.g., 'What is pi?'), quietly prompt: 'That's a start — can you make it more specific? What about pi puzzles you most?' 3. Collect notes and post them on the DQB yourself, grouping as you go and narrating: 'Waweru asks why some decimals stop and some don't — that's in this cluster. Akinyi asks whether Kenyan engineers use π or just 22/7 — I'm putting that over here.' 4. Read 6–8 questions aloud. For each, ask: 'Who else had something close to this?' (show of hands). 5. Vote: 'Raise your hand for the question you most want answered.' Record top 3. Write them prominently. 6. Post the unit driving question. Say: 'This is the question that connects everything we will learn in this unit. Your questions and this question will guide us. We will revisit this board at the start of every lesson and at the end to see how much closer we are to an answer.' 7. Read 3–4 blue sticky notes aloud: 'Mwangi says architects designing circular water tanks in Kisumu; Fatuma says doctors measuring dosages; Omondi says GPS satellites...' Say: 'These are exactly the people who need to understand real numbers. You will meet some of their challenges in this unit.'	signals that both intellectual curiosity and practical application are valued equally in this classroom. Keeping the DQB visible throughout the unit allows learners to experience their own growing understanding as questions get answered — a powerful metacognitive tool aligned with CBE's emphasis on self-directed learning.	direct indicator of depth of engagement with the phenomenon. (2) Do blue notes reference genuinely diverse Kenyan professional contexts, or do all learners copy one another? Diversity of contexts indicates breadth of transfer. (3) Can the learner articulate in the vote why they chose their top question — i.e., what about that question matters to them? Teacher photographs the DQB for their records and notes which question clusters dominate (this informs the sequencing emphasis for Lessons 2–6).
Model Building Phase  VIDEO: Compare rational numbers using a number line	Learners open a fresh double page in their exercise books labelled 'My Number Landscape — Unit Model (Start)'. They draw a large oval/blob shape representing 'All the numbers I know'. Inside the blob, they place the 10 numbers	1. Say: 'Scientists and mathematicians do not just collect facts — they build models. A model is a picture of how you currently understand something. Your model today will be incomplete. That is not a problem	Cumulative Model Revision Protocol (Model Building Phase): The explicit instruction that 'this model will be wrong and that is fine' is pedagogically critical — it removes the fear of incorrect initial models and frames revision as the	Observable indicators: (1) Does the learner spontaneously create sub-regions within their blob (e.g., separating whole numbers from decimals, or 'normal' numbers from π and $\sqrt{2}$)? This indicates readiness for formal classification

Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Properties of Rational Numbers versus Irrational Numbers (Flexbook) Source: CK-12 Search ARES for similar readings	from today's Number Sort Card wherever feels right — they can cluster them, spread them, or create sub-regions if they wish. Below their diagram, they write three sentences: (1) 'Right now, I think numbers can be divided into ___ types because...'; (2) 'The circumference of mama mboga's sufuria ($28 \times \pi$) belongs in my diagram at... because...'; (3) 'The biggest question my diagram cannot yet answer is...'. Learners are told explicitly: 'This is your starting model. It will be wrong in some ways — that is completely fine and expected. In every lesson we will return to this page and add, move, or revise things. By Lesson 8, your model will look very different from today's version.' Learners share their model with a partner for 2 minutes, explaining one thing they are confident about and one thing they are unsure of.	— it is the starting point.' 2. Give 6 minutes for individual model-drawing. Play soft background music (optional) to signal focused work time. Circulate and look for: learners who create sub-regions spontaneously (strong mathematical intuition); learners who place all 10 numbers in one undifferentiated blob (need scaffolding in Lesson 2). 3. After drawing, give 3 minutes for the three written sentences. Circulate and read sentence 3 ('biggest question my diagram cannot yet answer') — these feed directly into the DQB and inform lesson planning. 4. Signal partner share: 'Turn to your partner. You have 2 minutes. Person on the left goes first. Tell your partner one thing you are confident about in your model and one thing your model cannot yet explain.' 5. After partner share, cold-call 2 pairs: 'Njeri, what was your partner confident about?' WAIT 10 seconds. 6. Close the lesson by pointing to the DQB and the two-region diagram on the board: 'These two things — your questions on the board and your model in your book — are the most important things we produced today. Everything we learn in this unit will be building toward being able to answer those questions and complete that model. Mama mboga's sufuria will be with us the whole way.' 7. Preview Lesson 2: 'Next lesson, we will give names to the two regions in our model. Those names have been used by mathematicians for over 2,000 years. And we will find out whether $22/7$ really equals π — or just pretends to.'	goal rather than a sign of failure. By anchoring the model to the sufuria phenomenon from the first entry, every subsequent model revision will be tied to a concrete, memorable Kenyan context. The three structured sentences force learners to articulate current understanding, make a claim about the phenomenon, and identify the frontier of their own knowledge — the three cognitive moves that drive inquiry-based learning.	in Lesson 2. (2) Is the learner's sentence 2 (where does $28 \times \pi$ belong?) consistent with their Number Sort Card placement from Phase 1, or has their thinking shifted? A shift indicates genuine learning within the lesson. (3) Is sentence 3 (biggest unanswered question) mathematically substantive? Teacher collects exercise books or photographs sentence 3 from 6 representative learners (2 high / 2 mid / 2 developing) to inform the opening activity of Lesson 2.
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record your responses in your professional journal:

1. PHENOMENON ENGAGEMENT: Did the Ugali Pot Problem story create genuine curiosity, or did learners seem disengaged? If disengaged, what adjustment to the story or delivery might make it more vivid or locally relevant for your specific class?
2. PREDICT PHASE DATA: Which of the 10 numbers on the Sort Card generated the most disagreement? Were the misconceptions you observed consistent with what the lesson anticipated ($1/3$ vs. $0.333\dots$; $22/7$ vs. π)? Note the specific names of learners who showed strong proto-understanding of the terminating/repeating distinction — plan to use them as peer-explainers in Lesson 2.
3. DQB QUALITY: Review the yellow sticky notes. Are the learner questions rich enough to sustain a unit of inquiry, or are they mostly surface-level? If mostly surface-level, consider adding a 'question stems' scaffold in Lesson 2 (e.g., 'Why does... / What would happen if... / Is it always true that...').
4. MODEL QUALITY: Review the 6 photographed sentence 3 responses ('biggest question my model cannot yet answer'). Do these questions align with the Lesson 2 content (naming rational and irrational numbers) or do they reveal a different gap that needs to be addressed first?
5. CBE COMPETENCY REFLECTION: Communication and Collaboration — did all learners participate verbally in at least one phase today, or were some silent throughout? Note the names of silent learners and plan a deliberate cold-call or pair-share role assignment for Lesson 2 to ensure their voices are heard.
6. TIMING: Note which phase ran over time and which ran short. The Explain phase (Phase 3) typically runs long due to the gallery walk debate. If time is tight, the gallery walk can be reduced to a 2-minute silent read without sticky dots, with disagreements surfaced verbally instead.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	A cylindrical aluminium sufuria with a diameter of 28 cm from Gikomba Market; the calculator result of $28 \times \pi = 87.96459430\dots$ with digits that never end or repeat; the contrast between $28 \times 3 = 84$ (exact) and $28 \times \pi = 87.96459430\dots$ (inexact); a visual scroll of the digits of π continuing without pattern.
What did I learn?	Some numbers — like 60.50, 160, 34.5 — can be written down completely and exactly, while others — like π and $\sqrt{2}$ — cannot be written as terminating or repeating decimals; the circumference of a circle with a whole-number diameter may not itself be a 'nice' number; mathematicians classify numbers into different types based on whether they can be expressed exactly.
How does this explain the phenomenon?	We cannot yet fully explain why multiplying a whole number (28) by π produces a number that goes on forever without repeating, or what mathematical property separates numbers like π from numbers like $1/3$ or 60.50; we cannot yet explain whether $22/7$ is the same as π or merely close to it; we cannot yet name the two categories of numbers we have identified, or describe the full system of real numbers.

LESSON 2 (80 minutes): Sorting the Number World: Classifying Real Numbers from Kenyan Contexts

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners classify numbers encountered in Kenyan daily life into the formal categories of the real number system — natural numbers, whole numbers, integers, rational numbers, and irrational numbers — and represent this hierarchy visually, building the conceptual foundation needed to explain why π produces an endless, non-repeating decimal when calculating the circumference of the mama mboga's sufuria.
Knowledge	Learners will know: (1) the definitions and distinguishing features of natural numbers, whole numbers, integers, rational numbers, and irrational numbers; (2) that the real number system is structured as nested sets: $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$, with irrational numbers forming the complement of $\mathbb{Q}$ within $\mathbb{R}$; (3) that rational numbers can always be expressed as a ratio p/q ($p, q \in \mathbb{Z}, q \neq 0$) and produce terminating or repeating decimals; (4) that irrational numbers — such as π and $\sqrt{2}$ — cannot be expressed as such a ratio and produce non-terminating, non-repeating decimals.
Skills	Learners will be able to: (1) sort a given set of numbers into the correct subset(s) of the real number system; (2) justify their classification using mathematical reasoning; (3) represent the real number hierarchy using a nested Venn diagram; (4) place numbers accurately on a number line that includes integers, fractions, and irrational values.
Attitudes	Learners will appreciate that mathematical classification is a powerful tool for making sense of the world, value precision and justification in mathematical reasoning, and show curiosity about why some numbers — like the circumference of the sufuria — cannot be written down completely.
Key Inquiry Question	What are real numbers, and how do we classify the numbers we use in day-to-day situations in Kenya?
Purpose in Storyline	In Lesson 1, learners encountered the Ugali Pot Problem and recorded initial predictions about π . In this lesson (Lesson 2 of 8), learners formally encounter the full real number system for the first time. By sorting 20 Kenyan number cards — drawn directly from the phenomenon and from familiar Kenyan life — into self-invented and then formally guided categories, learners build the conceptual vocabulary needed to precisely answer WHY the circumference of the sufuria cannot be written exactly. The lesson opens the Driving Question Board (DQB) and positions every subsequent lesson as an investigation into a specific region of the real number system.
Safety Notes	No significant safety concerns for this lesson. Ensure learners handle printed number cards carefully. If scissors are used to cut printed cards, supervise use and collect scissors after the activity.

B. LESSON OVERVIEW

Learners are re-anchored to the Ugali Pot Problem: the mama mboga's daughter multiplied $28 \times \pi$ and got 87.96459430... — a decimal that never ends and never repeats. The lesson asks: what KIND of number is π , and how does it relate to numbers like Ksh 60.50, -7°C , or $2/3$ kg of unga? Working in pairs, learners receive 20 laminated Kenyan number cards and sort them using their own invented categories (Predict + early Observe). Groups share their sorting rules. The teacher then introduces the formal real number classification — $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$, with irrationals — using a projected nested Venn diagram (Explain). Learners reclassify their cards against the formal system and place selected numbers on a class number line (Observe + Explain continued). The Driving Question Board is formally opened: each pair posts one new question that emerged from the classification activity (DQB Creation). The lesson closes with learners sketching their first individual real number hierarchy model and annotating it with at least three of their Kenyan number cards (Model Building).

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Worked example: classifying numbers Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Classify numbers Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners are shown the phenomenon slide: the mama mboga's daughter's calculator screen displaying $28 \times \pi = 87.96459430\dots$ alongside four other Kenyan numbers: Ksh 60.50 (price of 1 kg of unga), 160 km (Nairobi to Nakuru), -7 (overnight temperature on Mt. Kenya summit in $^{\circ}\text{C}$), and $2/3$ (fraction of a sufuria of githeri eaten). Each learner individually writes answers to three questions in their exercise books: (1) 'Write down what TYPE of number each of the five numbers is — use whatever words you already know.' (2) 'Which of these five numbers can you write down completely and exactly? Circle them.' (3) 'Is π a fraction? Yes / No / Not sure — explain your thinking in one sentence.' Learners work silently for 4 minutes. No sharing yet.	Display the phenomenon slide and read aloud: 'Remember our mama mboga in Gikomba? Her daughter's calculator showed 87.96459430... when she multiplied 28 by π . Look at these five numbers on the board — they all come from Kenyan daily life.' Point to each number in turn. Say: 'Before we do anything else, I want to know what YOU think. In your books, answer these three questions on your own. Do not discuss — this is your honest prediction.' Set a 4-minute timer visibly. Circulate silently, scanning written responses. Do NOT correct or comment. After 4 minutes, say: 'Keep your books open to this page — we will come back to check your predictions.' Cold-call 3 learners (one strong, one mid, one hesitant) to read ONLY their answer to question 3. After each, say: 'Thank you — hold that thought.' Allow 10 seconds of wait time after each cold-call before moving on. Record the three responses as a tally on the board: π IS a fraction / π is NOT a fraction / Not sure.	Individual written prediction (Predict-Before-Instruction). By committing predictions to paper before any instruction, learners activate prior knowledge, expose misconceptions about number types, and create a personal baseline they will revise throughout the lesson. This also gives the teacher immediate formative data on the range of prior understanding in the room.	Observable indicator: Each learner has written a response to all three questions within 4 minutes. Teacher scans for: (a) learners who can already use terms like 'fraction', 'decimal', 'whole number', or 'integer' — these learners can be used as pair anchors; (b) learners who describe numbers only by appearance ('it has a decimal point') rather than by mathematical property — flag for targeted questioning during card sort. The tally of π IS / IS NOT / Not sure gives a whole-class snapshot of prior knowledge about irrationality.
Observe Phase  VIDEO: Classifying numbers Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Missing numbers	Each pair receives a set of 20 laminated Kenyan Number Cards (printed and cut before the lesson). The cards include: [Natural/Whole] 1, 7, 160, 28; [Integers — negative] -3 , -7 , -15 ; [Fractions/Rationals — fractional form] $2/3$, $3/4$, $7/2$; [Rationals — terminating decimal] 60.50, 34.5, 0.25; [Rationals — repeating decimal] 0.333..., 0.142857142857...; [Irrationals] π ,	Distribute card sets and say: 'You and your partner have 20 number cards. Every number on these cards comes from a real Kenyan context — prices, distances, temperatures, measurements. Your job is to sort them into groups. You choose the groups — there is no single right answer yet. You must name each group with a sticky note.' Start the 8-minute timer. Circulate actively. At each	Card Sort with Learner-Generated Categories (Elicit-Before-Tell). Learners construct their own classification schemes before the formal system is introduced. This surfaces the informal, intuitive understanding they bring (e.g., 'numbers with decimals' vs. 'whole numbers') and creates productive cognitive dissonance when the formal system challenges or refines their groupings. The	Observable indicators: (1) All pairs have produced at least 2 named groups within 8 minutes. (2) Pairs who have separated π and $\sqrt{2}$ from the decimal numbers show emerging awareness of the rational/irrational distinction — teacher notes these pairs for targeted questioning in Phase 3. (3) Pairs who have placed 0.333... with π show a misconception (confusing non-terminating with

on the number line Source: Khan Academy (English - US curriculum) Search ARES for similar readings	$\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$; [Zero] 0. Context labels are printed on each card: e.g., '160 km — distance Nairobi to Nakuru', '34.5°C — temperature in Kisumu', '0.333... — one-third kg of unga', 'π — used to find the circumference of the sufuria', '$\sqrt{2}$ — diagonal of a 1m $\times$ 1m farm plot in Kericho'. Pairs are given 8 minutes to sort the 20 cards into groups using ANY categories they choose. They must write their group names on sticky notes and place them above their groups. After 8 minutes, three pairs share their category names (not justifications yet) with the class by reading their sticky notes aloud. Teacher records all category names on the board without evaluating them.	pair, crouch to eye level and ask one probing question — choose from: 'Why did you put π and $\sqrt{2}$ together?', 'What makes -7 different from 7 in your system?', 'Where does 0.333... go — and why?', 'Is 160 the same kind of number as 160.0?' Do NOT correct; only probe. After 8 minutes, cold-call 3 pairs to read their sticky-note category names. Record every category name on the board exactly as stated. Say: 'Fascinating — we have [number] different ways of grouping these numbers. Let us keep them visible. Now I am going to show you the system that mathematicians use worldwide — including Kenyan engineers, doctors, and economists.'	Kenyan context on every card maintains direct connection to the anchoring phenomenon.	non-repeating) — teacher prioritises these pairs for the 'repeating vs. non-repeating' explanation in Phase 3. (4) Placement of 0 and negative numbers reveals understanding of integers and whole numbers.
Explain Phase  VIDEO: Proof: $\sqrt{2}$ is irrational Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Square Roots and Irrational Numbers (Flexbook) Source: CK-12 Search ARES for similar readings	The teacher projects a large nested Venn diagram of the real number system: $\mathbb{N}$ (natural numbers) $\subset$ $\mathbb{Z}$ (integers) $\subset$ $\mathbb{Q}$ (rational numbers) $\subset$ $\mathbb{R}$ (real numbers), with irrational numbers shown as the region inside $\mathbb{R}$ but outside $\mathbb{Q}$. Each region is colour-coded and labelled in both English and with a Kenyan example. The teacher walks through each set with its formal definition and two Kenyan examples. Learners then re-examine their 20 cards and, in pairs, re-sort them according to the formal system, placing each card in the correct region of a paper Venn diagram drawn in their exercise books. For each of four nominated cards (π, 0.333..., -7, 160), pairs must write a one-sentence justification: 'This number belongs in [set] because...'. Pairs then share justifications for π specifically — connecting directly back to the	Project the nested Venn diagram. Say: 'Mathematicians have a precise system for classifying ALL numbers. Let me walk you through it — and I want you to check whether your sorting categories match any of these official names.' Walk through each set: NATURAL NUMBERS — say: 'These are the counting numbers: 1, 2, 3... In Kenya, when a mama mboga counts her customers — 1, 2, 3 — she is using natural numbers. Which of your cards belong here?' [Wait 10 seconds. Cold-call 1 learner.] INTEGERS — say: 'Integers include all natural numbers, zero, AND negative whole numbers. The temperature on Mt. Kenya at night: -7°C. Which cards are integers but NOT natural numbers?' [Wait 10 seconds. Cold-call 1 learner.] RATIONAL NUMBERS — say: 'A rational number is any number that can be written as a fraction p over	Formal Definition Anchored to Phenomenon (Concept-Mapping + Justify-in-Writing). The formal classification is introduced AFTER learners have already attempted their own sorting, so the definitions land on fertile cognitive ground. Requiring written justifications ('This number belongs in [set] because...') forces learners to connect the formal definition to the specific number, rather than guessing by pattern-matching. Returning to the sufuria and π at the end of the phase closes the loop between abstract classification and the anchoring phenomenon.	Observable indicators: (1) Learners correctly place all 20 cards in their paper Venn diagrams — spot-check 5 pairs' diagrams during the 7-minute work period. Key error to watch for: placing 0.333... with π (irrational) rather than $\mathbb{Q}$ (rational). (2) Written justification for π must include BOTH criteria: 'cannot be written as p/q' AND 'decimal never terminates and never repeats.' Accept either criterion alone as 'Approaching Expectations'; require both for 'Meets Expectations.' (3) Thumbs response after each cold-call justification gives real-time whole-class comprehension data. If more than one-third of learners show thumbs sideways or down after the π justification, pause and re-teach using the calculator output 87.96459430... directly.

	sufuria phenomenon. The class collectively constructs the answer: 'π is irrational because it cannot be written as p/q, and its decimal expansion — 3.14159... — never terminates and never repeats, which is why the daughter could never cut the string to an exact length.'	q, where p and q are integers and q is not zero. This includes ALL terminating decimals — like Ksh 60.50 — AND all REPEATING decimals — like 0.333... Pause: is 0.333... a rational number?' [Wait 15 seconds. Cold-call 2 learners with opposing answers if available. Reveal: $0.333... = 1/3$, so YES, it is rational.] IRRATIONAL NUMBERS — say: 'Now here is the crucial one. An irrational number CANNOT be written as p/q. Its decimal expansion goes on FOREVER and NEVER repeats. π is 3.14159265358979... — it never repeats. $\sqrt{2}$ is 1.41421356... — it never repeats. These numbers live OUTSIDE the rational region but still INSIDE the real number system.' Point to the sufuria image and say: 'This is exactly why the daughter could never cut the string to a perfect length. π is irrational.' Give pairs 7 minutes to re-sort cards onto their paper Venn diagrams and write the four justification sentences. Then cold-call 4 pairs — one per nominated card — to read their justification aloud. After each, ask the class: 'Do we agree? Thumbs up, thumbs sideways, or thumbs down.' Address any disagreements immediately with a direct counter-example or proof sketch.		
Driving Question Board (DQB) Creation  VIDEO: Proof: square roots of prime numbers are irrational Source: Khan Academy (English - US curriculum)	Each pair is given two sticky notes. On sticky note 1, they write ONE new question that the classification activity raised for them — something they are now curious about or confused by. On sticky note 2, they write ONE thing they now understand more clearly about the sufuria problem than they did at the start of the lesson. Pairs post both sticky notes on the	Distribute two sticky notes per pair. Say: 'You have just classified 20 numbers. I want to know: what new questions does this raise for you? On your FIRST sticky note, write one question you now have — about any of the numbers, or about how we use them. On your SECOND sticky note, write one thing you now understand more clearly about the mama mboga's	Driving Question Board — Question Generation and Roadmapping. The DQB serves two simultaneous sensemaking functions: (1) it externalises and validates learner curiosity, signalling that questions are a legitimate and valued product of a mathematics lesson; (2) it gives the teacher a real-time, public record of learners' conceptual	Observable indicators: (1) Every pair has posted at least one question sticky note within the time limit — a pair with no question is a flag for confusion or disengagement; teacher follows up individually. (2) The QUALITY of questions reveals depth of understanding: questions about surface features ('What does the symbol $\square$ mean?') vs. questions

Search ARES for similar videos EXERCISE: Square Roots and Irrational Numbers Practice Source: CK-12 Search ARES for similar readings	class Driving Question Board, which is divided into four columns: 'Questions about Rational Numbers', 'Questions about Irrational Numbers', 'Questions about How We Calculate with These Numbers', and 'Other Questions'. The teacher reads 5–6 of the posted questions aloud and clusters related questions together. The teacher explicitly marks two questions as 'We will answer this in Lesson 3' and 'We will answer this in Lesson 5', showing learners the roadmap ahead. The board remains displayed for the full 8-lesson sequence.	sufuria problem than you did 80 minutes ago.' Give 3 minutes of silent writing time. Then say: 'Come and post your sticky notes in the column that fits best. If you are not sure which column, post it in Other Questions.' Allow 2 minutes for posting. Return to the front and read 5–6 questions aloud — select questions that preview Lessons 3–8 (e.g., questions about converting fractions to decimals, about how to calculate with π, about why $\sqrt{2}$ is irrational, about rounding). After reading each selected question, say: 'This is an excellent question — write it large on the board under the relevant column.' Mark two questions with a star and say: 'We will specifically investigate THIS question in Lesson 3' and 'THIS question in Lesson 5.' Say to the class: 'This board will grow every lesson. By Lesson 8, we will have answered every question on it.'	gaps and emerging interests, which directly informs the design of Lessons 3–8. Publicly mapping posted questions to future lessons builds learner agency and metacognitive awareness of their own learning trajectory.	about structural properties ('Can an irrational number ever be negative?') indicate different levels of conceptual engagement. (3) The 'understanding' sticky notes are collected by the teacher at the end of the lesson and reviewed to assess whether the lesson's core SLO (classify numbers into the real number hierarchy) has been achieved by the majority of learners.
Model Building Phase VIDEO: Compare rational numbers using a number line Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Definite integrals: common functions Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Each learner individually draws their first Real Number System Model in their exercise book. The model must include: (1) a nested Venn diagram with all five labelled regions ($\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, Irrationals, $\mathbb{R}$); (2) at least six of the 20 Kenyan number cards placed in the correct regions of the diagram (learners choose which six); (3) one annotation connecting the model back to the sufuria: 'π belongs here because...'; (4) a title: 'My Real Number System — Lesson 2.' Learners are told: 'This model is NOT finished. You will add to it in every lesson. By Lesson 8, it will show everything we have learned about real numbers.' The teacher circulates and initials 8–10 models to confirm correct structure before	Say: 'For the last 8 minutes, I want each of you to build your own model of the real number system. Open your exercise book to a fresh double page. Draw a large nested Venn diagram — big enough to write in each region. Label all five regions with their correct mathematical names and symbols. Then choose any SIX of your number cards and write them in the correct region. You MUST include π and write one sentence explaining why it is in the irrational region — connect it back to the sufuria problem.' Start the timer. Circulate and initial models that are correctly structured. If a learner has π in the wrong region, crouch and ask quietly: 'Can π be written as p divided by q where	Cumulative Individual Model Building (Progressive Model Revision). By building a personal model at the end of every lesson, learners create a visual, cumulative record of their growing understanding of the real number system. The requirement to place Kenyan number cards — including π — inside the model in every lesson maintains the direct connection to the anchoring phenomenon and prevents the model from becoming an abstract diagram divorced from the original problem. Initialling correct models during circulation provides immediate, low-stakes formative feedback.	Observable indicators: (1) The nested structure is correct — $\mathbb{N}$ inside $\square$ inside $\square$ inside $\square$, with irrationals in the $\square$-but-not-$\square$ region. A flat (non-nested) diagram indicates the learner has not yet grasped the subset relationship. (2) π is correctly placed in the irrational region with a written justification that references either 'cannot write as p/q' or 'decimal never terminates and never repeats.' (3) All six chosen numbers are in their correct regions — errors on 0, negative numbers, or repeating decimals are the most common and should be prioritised in Lesson 3 review. Teacher reviews all un-initialled books before Lesson 3 and prepares a 3-minute 'model

	the lesson ends.	both p and q are integers? Think about the calculator screen — 87.96459430... What do you notice about those digits?' [Wait 10 seconds before offering further guidance.] With 2 minutes remaining, say: 'Write your title at the top: My Real Number System — Lesson 2. You will add to this model in every lesson. Do not close this page.' Collect exercise books from any learner whose model you have NOT initialled, for after-class review.		correction' warm-up if more than 25% of learners show structural errors.
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D. TEACHER REFLECTION
After the lesson, reflect on the following questions: (1) During the card sort (Phase 2), what were the most common learner-generated category names? Did any pairs spontaneously use the terms 'rational' or 'irrational' before instruction? This reveals the extent to which learners bring prior vocabulary from Junior School. (2) How many pairs placed 0.333... in the irrational group during the initial sort? This is the most critical misconception in this lesson (confusing non-terminating with non-repeating). If more than one-third of pairs made this error, plan a 5-minute dedicated warm-up for Lesson 3 using the long division of $1 \div 3$ to show the repeating block. (3) Review the DQB sticky notes: do learners' questions cluster around rational numbers, irrational numbers, or operations? This should directly shape the emphasis of Lessons 3 and 4. (4) Review the 'understanding' sticky notes collected at the end of Phase 4: can learners articulate, in their own words, why the sufuria circumference cannot be written exactly? If fewer than 60% can do so, the phenomenon connection needs to be re-emphasised at the start of Lesson 3. (5) From the initialled models (Phase 5): what percentage of learners produced a correctly nested Venn diagram on the first attempt? If fewer than 70% did so, consider providing a partially pre-drawn template in Lesson 3 rather than requiring learners to construct the diagram from scratch.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners sorted 20 Kenyan number cards — including prices in Kenyan shillings, distances in km, temperatures in °C, fractions, repeating decimals, and irrational numbers such as π and $\sqrt{2}$ — into self-invented groups, then re-sorted them according to the formal real number classification system using a projected nested Venn diagram.
What did I learn?	The real number system is organised as nested sets: natural numbers $\subset$ integers $\subset$ rational numbers $\subset$ real numbers, with irrational numbers occupying the region inside $\square$ but outside $\square$. Rational numbers can always be written as p/q and produce terminating or repeating decimals; irrational numbers — such as π and $\sqrt{2}$ — cannot be written as p/q and produce non-terminating, non-repeating decimals.
How does this explain the phenomenon?	π is an irrational number. Because it cannot be expressed as a fraction p/q , the circumference of the mama mboga's sufuria (28π cm) cannot be written as an exact decimal — the digits 87.96459430... continue forever without repeating. This is why the daughter's string always has a tiny gap or overlap no matter how many decimal places she uses.

LESSON 3 (80 minutes): Lesson 3 — "Rational or Irrational? Sorting the Number Family"

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To deepen learners' ability to classify real numbers as rational or irrational, distinguish between terminating, recurring and non-terminating non-recurring decimals, and connect this classification to the Ugali Pot Problem phenomenon — understanding precisely why π produces a circumference that can never be written as an exact fraction.
Knowledge	Learners will know that: (i) rational numbers are numbers expressible in the form p/q where p, q are integers and $q \neq 0$; (ii) irrational numbers cannot be expressed as such a fraction and their decimal expansions neither terminate nor recur; (iii) together, rational and irrational numbers form the complete set of real numbers; (iv) common irrational numbers include π , $\sqrt{2}$, $\sqrt{3}$, and e ; (v) everyday measurements (prices in Ksh, distances in km, temperatures) are typically rational, while geometric constants such as π arise from irrational relationships.
Skills	Learners will be able to: (i) classify a given number as rational or irrational with justification; (ii) convert a terminating or recurring decimal to a fraction; (iii) place rational and irrational numbers correctly on a number line; (iv) use a calculator to observe the decimal behaviour of numbers and draw conclusions about their classification; (v) construct a counter-example to disprove a false classification claim.
Attitudes	Learners will appreciate that mathematical precision matters in real-life Kenyan professional contexts — from engineers designing water tanks in Turkana to architects drawing circular foundations in Nairobi — and that working confidently with irrational numbers requires understanding their nature rather than avoiding them.
Key Inquiry Question	What are real numbers? How do we use real numbers in day-to-day situations?
Purpose in Storyline	In Lesson 1 learners were introduced to the Ugali Pot Problem and confronted the puzzle that $28 \times \pi$ produces a never-ending, non-repeating decimal. In Lesson 2 learners placed various real numbers on the number line and explored their representation. Lesson 3 now gives learners the formal classification tools to answer the puzzle directly: they investigate the decimal structure of many everyday numbers — unga prices, distances, square roots of areas — and discover that most are rational (their decimals stop or repeat), while π and $\sqrt{2}$ are provably irrational. By the end of this lesson learners can explain precisely why the sufuria circumference can never be written as an exact fraction, advancing the driving question significantly.
Safety Notes	No hazardous materials are used. If physical string and a sufuria are brought to class, ensure the sufuria edges are not sharp. Calculators and mobile phones used for computation should be handled responsibly. No safety concerns beyond standard classroom practice.

B. LESSON OVERVIEW

Learners begin by predicting whether a set of ten Kenyan everyday numbers are rational or irrational before any instruction. They then carry out a structured calculator investigation — observing the decimal expansions of numbers drawn from Kenyan market prices, distances, temperatures, and geometric constants — and sort them into a classification table. In the Explain phase the teacher formalises the definition of rational and irrational numbers using the p/q criterion, demonstrates the recurring-decimal-to-fraction conversion technique, and connects these ideas directly back to why π produces the gap in the sufuria string. Learners update the Driving Question Board with new evidence, and close by revising their cumulative real-number model to show the rational/irrational partition inside the real-number set.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Proof: square roots of prime numbers are irrational Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Square Roots and Irrational Numbers (Flexbook) Source: CK-12  Search ARES for similar readings	Each learner receives a card strip listing ten numbers drawn from Kenyan everyday life: (1) 60.5 — price of 1 kg unga in Ksh; (2) 160 — distance Nairobi to Nakuru in km; (3) 34.5 — temperature in Kisumu in °C; (4) 0.333... — one-third of a 3-litre cooking-oil bottle; (5) $\sqrt{4}$ — side of a square plot whose area is 4 m ² ; (6) $\sqrt{2}$ — diagonal of a 1 m × 1 m paving tile; (7) π — ratio of sufuria circumference to diameter; (8) 22/7 — the fraction a carpenter uses to approximate π ; (9) 0.142857142857... — $1 \div 7$ on a calculator; (10) $\sqrt{5}$ — a measurement arising in a right-triangle roof truss design. Learners work individually for 5 minutes, sorting each number into one of three columns in their exercise books: 'Definitely can write as a fraction', 'Definitely CANNOT write as a fraction', 'Not sure'. Learners then share predictions with a shoulder partner for 2 minutes. No feedback is given yet.	Distribute card strips as learners settle. Say: 'Before we do any mathematics together, I want to know what you already think. Look at these ten numbers — all of them appear in Kenyan everyday life. Sort them silently into the three columns. You have 5 minutes. Go.' Circulate and observe — do NOT correct. After 5 minutes say: 'Now share your sorting with your partner. Convince them where you placed each number and why.' After pair-share, cold-call 4 learners (choose one who placed $\sqrt{2}$ as 'fraction', one who placed 0.333... as 'cannot', one uncertain about 22/7, one uncertain about 0.142857...). Record their predictions visibly on the board in a class tally. Say: 'We are not judging right or wrong yet. By the end of today you will be able to defend every one of these placements with mathematical evidence.' Leave predictions visible throughout the lesson.	Predict-then-Investigate (Elicitation of Prior Knowledge): surfacing misconceptions before instruction allows the teacher to target instruction precisely. Learners who incorrectly place 0.333... as 'cannot write as fraction' reveal a gap between recognising fractions and recognising recurring decimals as fractions — a key sensemaking moment to address in Phase 3.	Teacher observes sorting patterns during circulation. Key indicators: (a) most learners correctly identify 60.5, 160, 34.5 as rational — if not, prior number-sense needs revisiting; (b) learners who place 22/7 and π in the same column reveal a crucial misconception to address; (c) the proportion of 'not sure' responses for $\sqrt{2}$ and $\sqrt{5}$ signals readiness level for the irrational number concept.
Observe Phase  VIDEO: Proof: $\sqrt{2}$ is irrational Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Square Roots and Irrational Numbers Practice Source: CK-12  Search ARES for similar readings	Learners work in pairs. Each pair has a scientific calculator (or a mobile phone calculator). They receive a structured observation worksheet titled 'Decimal Detective'. For each of the ten numbers on their card strip they: (a) enter the number into the calculator and record the full display; (b) describe the decimal pattern in one sentence (e.g., 'stops after 1 decimal place', 'same 6 digits repeat forever', 'never stops, no pattern I can see'); (c) attempt to write the number as a fraction p/q — or write 'I cannot find any fraction that equals this	Say: 'You are now going to be decimal detectives. Your job is to look very carefully at what your calculator shows and describe what you see — like a scientist recording an observation, not jumping to conclusions yet.' Allow 15 minutes for calculator investigation. Circulate — pause at pairs who may think $22/7 = \pi$ and ask: 'Enter $22 \div 7$ on your calculator. Now enter π directly. Look at the digits after 3.14 — are they identical?' [WAIT 12 seconds.] Cold-call 2 pairs to share what they notice. Then project a large calculator display at	Data Collection and Pattern Recognition (NGSS Science and Engineering Practice 3 — Planning and Carrying Out Investigations; Practice 4 — Analysing and Interpreting Data): learners act as empirical investigators, using the calculator as a measurement instrument. The deliberate juxtaposition of 22/7 (repeating decimal, therefore rational) against π (non-repeating, therefore irrational) is the pivotal observational moment that directly answers why the sufuria string always has a gap — because no fraction can ever equal π exactly.	Check worksheets for: (a) correct decimal descriptions for all 10 numbers — especially that 0.142857142857... is described as 'repeating' not 'random'; (b) pairs who correctly attempt 1/3 for 0.333... are ready for the formal conversion technique; (c) pairs who write 'the repeating ones can be fractions, the non-repeating ones cannot' in their summary sentence have independently arrived at the core classification criterion — affirm this publicly.

	exactly'. For numbers 1–3 and 5 and 8 this is straightforward. For number 4 (0.333...) learners discover $\frac{1}{3}$ by reasoning. For numbers 6, 7, 10 learners observe the non-terminating non-repeating display and attempt — and fail — to find an exact fraction. The teacher then shows, using a projected calculator, that $\frac{22}{7} = 3.142857142857...$ (repeating) while $\pi = 3.14159265358979...$ (no repetition visible across all displayed digits). Learners record this critical difference in red pen in their worksheets. The observation closes with each pair writing one sentence: 'We think the key difference between Group A numbers and Group B numbers is...'	the front showing both: $\frac{22}{7} = 3.142857142857142857...$ versus $\pi = 3.14159265358979323846...$ Say: 'A Kenyan architect in Nairobi and a mama mboga in Gikomba both use $\frac{22}{7}$ as an approximation for π. Is $\frac{22}{7}$ actually equal to π?' [WAIT 10 seconds.] Cold-call 3 learners. Then say: 'Write the key difference you observe in red — this is the most important observation of today.' After pairs write their sentences, cold-call 3 pairs to share. Record key phrases on the board.		
Explain Phase  VIDEO: Compare rational numbers using a number line Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Compare rational numbers using a number line Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	The teacher delivers a structured explanation in three blocks, each followed by a learner practice task. BLOCK 1 (10 min) — Formal Definitions: The teacher writes the definition of a rational number ($\frac{p}{q}$, integers, $q \neq 0$) and an irrational number on the board. Learners copy and then immediately classify all 10 card-strip numbers using the formal definition, revising their Phase 1 predictions. BLOCK 2 (10 min) — Converting Recurring Decimals to Fractions: Teacher demonstrates the algebraic method using $0.333... \rightarrow \text{let } x = 0.333..., \text{ then } 10x = 3.333..., \text{ subtract: } 9x = 3, \text{ so } x = \frac{1}{3}$. Learners practise with $0.142857142857... (= \frac{1}{7})$ and $0.181818... (= \frac{2}{11})$. BLOCK 3 (8 min) — Back to the Phenomenon: Teacher explicitly connects to the sufuria problem. Learners write a 'phenomenon connection sentence': 'The circumference of	BLOCK 1: Write definition on board and say: 'A rational number is any number that can be written in the form $\frac{p}{q}$, where p and q are integers and q is not zero. An irrational number is a real number that CANNOT be written in this form — its decimal expansion never terminates and never repeats.' Point to the class prediction tally still on the board. Ask: 'Using this definition, look at your Phase 1 tally — who needs to move a number?' [WAIT 12 seconds.] Cold-call 3 learners to justify a reclassification. Target learner who placed $\sqrt{4}$ as irrational — ask: 'What is $\sqrt{4}$ exactly?' [WAIT 8 seconds.] Establish $\sqrt{4} = 2 = \frac{2}{1}$, therefore rational. BLOCK 2: Say: 'Now I am going to show you a mathematical trick that reveals why recurring decimals are rational — they ARE fractions in disguise.' Work through $0.333...$ on the board step by step. Then say:	Formal Definition Anchored to Prior Observation (NGSS Practice 6 — Constructing Explanations): learners move from empirical pattern observation (Phase 2) to formal mathematical language. The recurring-decimal-to-fraction conversion is a concrete procedural bridge that makes the abstract $\frac{p}{q}$ definition tangible. The phenomenon reconnection in Block 3 ensures the formal language serves the driving question rather than existing in isolation.	Exit check after Block 1: scan 6 learner books to confirm $\sqrt{4} = \text{rational}$ is now corrected. After Block 2: the board work by the cold-called learner for $0.142857...$ is visible to all — correct any algebraic errors publicly. After Block 3: listen to the three read-aloud sentences. A complete sentence must include: (i) the word 'irrational', (ii) a reference to the decimal expansion not terminating or repeating, and (iii) the implication for exact measurement. Sentences missing any element flag learners for targeted follow-up.

	the sufuria (28π cm) cannot be written as an exact fraction because π is irrational, which means ____.' They complete the sentence and share with their partner.	'Your turn. Work out whether 0.142857142857... is rational using the same method.' [WAIT 15 seconds for individual attempt.] Cold-call 1 learner to the board. Affirm and then say: 'This is exactly why your calculator shows $1 \div 7$ as a repeating decimal — it is rational.' BLOCK 3: Ask: 'So — back to Gikomba Market. The sufuria has a diameter of exactly 28 cm. The daughter multiplies by π. Why does she ALWAYS get a gap or overlap when she cuts the string?' [WAIT 12 seconds.] Cold-call 2 learners. Then say: 'Complete this sentence in your book: The circumference cannot be written as an exact fraction because...'. After 3 minutes, cold-call 3 learners to read their sentence aloud. Correct any conflation of π and $22/7$.		
Driving Question Board (DQB) Creation  VIDEO: Proof: there's an irrational number between any two rational numbers Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Rational number word problems Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners return to the class Driving Question Board. The DQB has three columns: 'What we observed', 'What we now understand', 'New questions we have'. Working in groups of four, learners write new sticky notes for each column based on today's lesson. Suggested contributions: 'Observed' — '$22/7$ and π look similar on a calculator but have different decimal structures'; 'Understand' — 'π is irrational because its decimal never terminates or repeats — so no string length can be exactly right'; 'New questions' — 'Are there other irrational numbers in Kenyan building and cooking? Can we prove $\sqrt{2}$ is irrational?' Groups post their notes and the teacher facilitates a 5-minute gallery walk where learners place a star sticker on the most insightful 'New	Say: 'It is time to update our Driving Question Board. Remember our question: why can we never write some everyday numbers down completely? You now have new evidence and new understanding. In your groups, write at least one sticky note for each column. You have 5 minutes.' Circulate and prompt groups who are stuck with: 'What was the most surprising thing your calculator showed you today?' After posting, say: 'Do a quick gallery walk — read the other groups' notes and place a star on the new question you find most mathematically interesting.' After the gallery walk, cold-call 2 learners to read the starred questions aloud. Acknowledge and say: 'These questions will drive Lessons 4 and 5 — we will investigate surds and the proof that $\sqrt{2}$ is irrational.'	Collaborative Knowledge Consolidation via DQB (NGSS Practice 7 — Engaging in Argument from Evidence; Practice 8 — Obtaining, Evaluating and Communicating Information): the DQB externalises the class's collective growing understanding. The gallery walk with star stickers creates peer evaluation of questions, building metacognitive awareness of what is still unknown — a key competence in the CBE framework.	Teacher reads all sticky notes during circulation. Quality indicators: 'Observed' notes should reference specific decimal evidence (not vague impressions); 'Understand' notes should use the word 'irrational' or 'rational' correctly; 'New question' notes reveal depth of curiosity — questions about proof of irrationality indicate high-level thinking. Photograph the board for reference in planning Lessons 4–5.

	question' note from another group.			
Model Building Phase  VIDEO: Worked example: classifying numbers Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Properties of Rational Numbers versus Irrational Numbers (Flexbook) Source: CK-12  Search ARES for similar readings	Learners retrieve the cumulative real-number model they began in Lesson 1 (a large set diagram in their exercise books). In Lesson 1 they drew the outer set 'Real Numbers'. In Lesson 2 they added a number line inside it. Today they add the key structural partition: they draw two non-overlapping subsets inside the Real Numbers set, label one 'Rational Numbers (Q)' and one 'Irrational Numbers'. Inside the Rational Numbers subset they place examples from today: 60.5, 160, 34.5, $\frac{1}{3}$, $\frac{22}{7}$, $0.142857... = \frac{1}{7}$, $\sqrt{4} = 2$. Inside the Irrational Numbers subset they place: π, $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$. They annotate each irrational number with a one-phrase note explaining its decimal behaviour. They also add a small inset sketch of the sufuria with the label '28π cm — irrational circumference' with an arrow pointing to the Irrational Numbers subset. Learners share their model with a partner and give one piece of feedback using the stem: 'I notice you placed ___ in _____. I agree/disagree because...'	Say: 'Open your cumulative model from Lessons 1 and 2. Today we are adding the most important structural feature of the real-number system. Draw two non-overlapping regions inside your Real Numbers set. Label them Rational Numbers — using the symbol Q — and Irrational Numbers. Now place every number from today's card strip into the correct region and annotate.' Allow 8 minutes. Circulate and look for: learners who leave $\sqrt{4}$ in Irrational Numbers (address immediately by asking 'What is the exact value of $\sqrt{4}$?'); learners who place π and $\frac{22}{7}$ in the same region (redirect to their Phase 2 calculator observation). After 8 minutes say: 'Share your model with your partner. Give feedback using the sentence stem on the board: I notice you placed blank in blank — I agree or disagree because blank.' Allow 4 minutes for peer feedback. Close by cold-calling 2 learners to describe their model to the class, emphasising the sufuria annotation. Say: 'Every lesson we will add one more layer to this model. By Lesson 8 it will be a complete map of the real number system — and it will answer our driving question fully.'	Cumulative Model Revision (NGSS Practice 2 — Developing and Using Models): the set diagram is a living scientific model that learners revise lesson by lesson. Adding the rational/irrational partition today is the single most structurally important revision in the 8-lesson sequence because it establishes the fundamental topology of the real number system. Anchoring the sufuria in the irrational subset makes the model a direct answer-in-progress to the driving question.	Collect 6 exercise books at random at the end of the lesson. Check: (a) the two subsets are drawn as non-overlapping and both inside Real Numbers; (b) all 10 card-strip numbers are placed correctly; (c) irrational numbers each carry a decimal-behaviour annotation; (d) the sufuria annotation is present. Any book missing (b) or (d) indicates a learner who has not yet connected the classification to the phenomenon — flag for one-on-one follow-up at the start of Lesson 4.

D. TEACHER REFLECTION

After the lesson, consider: (1) Did learners independently arrive at the repeating-vs-non-repeating distinction in Phase 2, or did most need the projected calculator comparison? If most needed the projection, design a more guided observation worksheet for the next cohort. (2) How many learners still conflated $\frac{22}{7}$ with π after the Block 3 explanation? This is a persistent and important misconception in Kenyan classrooms — if more than 25% of learners still conflate them, plan a brief 5-minute consolidation activity at the start of Lesson 4. (3) Did the QDB 'New Questions' column generate genuine curiosity about proof and surds? If questions were thin or repetitive, consider introducing a 'mathematical mystery card' stimulus in Lesson 4 — e.g., a short story about a Kenyan engineer who could not use an exact value for $\sqrt{2}$ when designing a diagonal roof beam. (4) Were learners able to give substantive peer feedback on the cumulative model, or was it superficial? If feedback was weak,

model a high-quality feedback exchange explicitly at the start of the next peer-review activity. (5) Note the names of learners whose Phase 1 predictions were most accurate — these learners may be strong peer-teachers for differentiated grouping in Lesson 4.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed the decimal expansions of ten Kenyan everyday numbers on a calculator and discovered that some decimals terminate (60.5), some repeat in a pattern ($0.142857142857\dots = 1/7$; $0.333\dots = 1/3$; $22/7 = 3.142857\dots$), and others — particularly π and $\sqrt{2}$ — neither terminate nor repeat across all visible digits.
What did I learn?	Learners learned that rational numbers are expressible as p/q (integers, $q \neq 0$), that terminating and recurring decimals are always rational, that irrational numbers have non-terminating non-repeating decimal expansions, and that π is irrational — which is exactly why multiplying the sufuria's diameter (28 cm) by π produces a circumference that can never be written as an exact fraction or a terminating decimal.
How does this explain the phenomenon?	The formal classification of real numbers into rational and irrational subsets, the algebraic technique for converting a recurring decimal to a fraction, and the direct connection between π 's irrationality and the unavoidable gap or overlap when cutting a string to fit the sufuria rim were explained and practised.

LESSON 4 (80 minutes): Placing Numbers on the Line: Can We Always Find the Exact Spot?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners physically construct a classroom number line and place rational and irrational numbers on it, justifying each position using decimal approximations and benchmark reasoning, thereby deepening their understanding of how all real numbers — including non-terminating, non-repeating ones — occupy precise (if unwriteable) locations on the number line.
Knowledge	Learners will know that: (1) every real number — rational or irrational — corresponds to exactly one point on the number line; (2) irrational numbers such as $\sqrt{2}$, $\sqrt{3}$ and π can be approximated to any desired degree of accuracy to locate them; (3) temperature readings and other everyday Kenyan measurements can be classified as rational or irrational; (4) the number line is a complete, continuous model of the real number system with no gaps.
Skills	Learners will be able to: (1) estimate the decimal value of simple surds ($\sqrt{2} \approx 1.414$, $\sqrt{3} \approx 1.732$) and expressions such as $\pi - 3 \approx 0.1416$ using a calculator or known benchmarks; (2) order a mixed set of real numbers and place them correctly on a scaled number line; (3) justify the placement of each number using mathematical reasoning; (4) classify a set of real-world Kericho temperature data as rational or irrational.
Attitudes	Learners will appreciate that: (1) mathematical precision and physical reality sometimes require careful approximation; (2) irrational numbers are not 'broken' or 'incomplete' — they have exact positions even when they cannot be written exactly; (3) collaborative reasoning and respectful debate sharpen mathematical justification skills.
Key Inquiry Question	Can we always find the exact spot for an irrational number on the number line — and what does that tell us about the kind of number the ugali-pot circumference really is?
Purpose in Storyline	In Lessons 1–3 learners identified and classified rational and irrational numbers and explored non-terminating decimals. In this lesson (4 of 8) learners make the abstract number line concrete: they stretch a real string across the classroom (mimicking the daughter's string around the sufuria rim) and must physically peg irrational numbers like $\sqrt{2}$ and $\pi - 3$ onto it. This bridges classification (what kind of number is it?) to representation (where exactly does it live?), directly answering why the daughter's string always has a tiny gap — 88 cm is a peg placed slightly off the true irrational position. The Kericho temperature subplot grounds the classification skill in a second real Kenyan context.
Safety Notes	When stretching the string number line across the classroom, ensure the string is held at a safe height (below neck level, approximately waist height) and is secured at both ends before learners move around it. Scissors used to cut number cards should be handled with care. Learners with mobility difficulties should be assigned card-holding or recording roles at a fixed station.

B. LESSON OVERVIEW

Learners collaborate to construct a large physical number line spanning -5 to 5 using string and pegs strung across the classroom. Working in groups, they receive eight number cards ($-\sqrt{3}$, -1.5 , 0 , $2/3$, $\sqrt{2}$, $\pi - 3$, $2.333\dots$, and 3) and must estimate, debate, and physically peg each card to its correct position on the line, providing a written justification. The lesson is anchored by returning to the Ugali Pot Problem: the daughter's string (88 cm) is itself a 'peg' placed near — but not exactly at — the irrational circumference. A second supporting phenomenon, the Kericho Tea-Country Temperature Record, is introduced: learners receive a table of seven daily temperatures from Kericho and must plot these on a mini number line strip, classifying each as rational or irrational. The lesson closes with a Driving Question Board update focused on the new question: "Can we always find the exact spot for an irrational number?"

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Plotting decimal numbers on a number line Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Negative decimals on the number line Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Each learner receives a slip showing the eight numbers: $-\sqrt{3}$, -1.5 , 0 , $\frac{2}{3}$, $\sqrt{2}$, $\pi - 3$, $2.333\dots$, and 3 . Without a calculator, learners individually sketch a mini number line from -5 to 5 in their exercise books and mark where they think each number belongs, writing a one-sentence justification for each placement. They then compare predictions with a partner, noting any disagreements before the whole-class share.	Display the eight number cards on the board in random order. Say: 'Before we touch any string or calculator, I want your best guess for where each of these numbers lives on a number line from -5 to 5 . Use what you already know — benchmarks, what you remember about π , square roots, fractions. Sketch it yourself first — 3 minutes, silent, pencils only.' Set a visible 3-minute timer. After 3 minutes: 'Now turn to your elbow partner. You have 90 seconds — find ONE number you disagree about and explain why.' After partner talk, cold-call 4 pairs (select one strong, one mid, two uncertain based on prior lesson exit tickets): 'Amina and Brian — which number did you disagree about and what were your two positions?' WAIT 12 seconds after each pair responds before probing. For each response ask: 'What number did you use as a benchmark to decide that?' Do NOT confirm or correct yet — record disagreements on the board in a 'We disagree about...' column. Close by returning to the phenomenon: 'Remember the daughter cutting string for the sufuria rim — she had to decide where to cut. Today we are doing exactly that, but with a number line instead of a pot.'	Benchmark Anchoring Prediction: learners use known reference points (0 , 1 , -1 , 2 , integers) as anchors to estimate unfamiliar values. This activates prior knowledge about the relative magnitude of surds and π before formal computation, and surfaces misconceptions (e.g. learners who place $\sqrt{2}$ at 2 or $\pi - 3$ near 0.5) that the Observe phase will address.	Circulate and photograph 3–4 learner mini number lines. Look for: (1) Does the learner correctly place 0 and integers as anchors? (2) Is $-\sqrt{3}$ placed to the left of -1 ? (3) Is $\pi - 3$ placed between 0 and 1 ? Flag learners who place all irrationals at integer positions — these learners need targeted support in Phase 3. Record names for cold-calling in Phase 3.
Observe Phase  VIDEO: Compare rational numbers using a number line	Groups of 4 receive: a set of 8 laminated number cards, a calculator, a 'Decimal Approximation Table' worksheet (two columns: Number Decimal approximation to 4 d.p.), and a	Before releasing groups: 'Your job in the next 18 minutes is two things. First: fill in the Decimal Approximation Table — every number to 4 decimal places. Second: peg your 8 cards onto the	Physical Embodied Modelling: the act of pegging a card to a real string makes the abstract density of the real number line tangible. Learners must commit to a physical location, creating	Check the Decimal Approximation Tables as you circulate: (1) Can learners correctly compute $\sqrt{2} \approx 1.4142$ and $\sqrt{3} \approx 1.7321$? (2) Do learners recognise $\pi - 3 \approx 0.1416$ (not 0.5 or 3.14)? (3) For the

Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Decimals on the number line: tenths Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Kericho Temperature Record card showing 7 days of temperatures (e.g. 18°C, 19.5°C, $\sqrt{5}$ °C above baseline [$\approx 2.236^\circ\text{C}$ above], 20°C, 18.75°C, 21°C, 19.333...°C). Groups first complete the Decimal Approximation Table for all 8 number-line cards using their calculators. They then examine the string number line (already stretched across the classroom, marked with tape at every integer from -5 to 5 and subdivided at every 0.5) and peg each card to what they believe is the correct position. Groups must also classify each of the 7 Kericho temperatures as rational or irrational and plot them on a printed mini number-line strip (16°C to 23°C). The teacher circulates but does not confirm placements — groups record their decimal approximations and initial peg positions on the worksheet.	class number line. But — here is the rule — before you peg, your group must agree on a justification sentence: WHY does this number go here? Write it on the back of the card.' Assign roles within each group: Calculator Operator, Peg Placer, Justification Writer, Kericho Temperature Analyst. Circulate every 3 minutes. At each group, ask ONE targeted question — do NOT answer questions about placement: 'You've placed $\pi - 3$ there — what decimal did your calculator give you? Where is that between on the number line?' and 'For $\sqrt{3}$ — is it bigger or smaller than 1.5? How do you know?' At the 10-minute mark, call attention: 'Kericho Temperature Analysts — have you found any irrational temperatures in that data? What makes you say that?' WAIT 10 seconds. At the 15-minute mark, ring a bell: 'Peg Placers — step back from the line. Look at the whole class line together. Do all groups agree on where 2.333... is sitting? Look carefully.' Allow 2 minutes of silent inspection before Phase 3.	productive cognitive dissonance when groups place the same number in slightly different positions. The Kericho temperature subplot uses a familiar Kenyan context (Kericho tea-country weather) to reinforce that the classification skill transfers beyond the sufuria.	Kericho data, do learners identify $\sqrt{5}$ above baseline as irrational and $18.75 = 75/4$ as rational? Flag any group that places 2.333... between 2 and 2.25 — this suggests they do not recognise $2.333... = 7/3 \approx 2.333$. Note this for the Explain phase. Take a photo of the full class number line before Phase 3 for reference.
Explain Phase VIDEO: Proof: square roots of prime numbers are irrational Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Round decimals using a number line Source: Khan	The class gathers around the string number line. Each group defends ONE assigned card's placement (teacher assigns so that stronger groups get the irrational numbers). The defending group reads their justification sentence aloud. The rest of the class acts as a 'Mathematical Court' — they may challenge the placement with evidence (a different decimal approximation, a benchmark argument) or support it. After all 8 cards are defended and finalised, learners complete a 'Number Type Gallery' in their books: a table with	Open Phase 3: 'We have a number line full of pegged cards. Some may be in the wrong place. We are going to hold a Mathematical Court. Each group will defend one card — then the court decides: keep it or move it, and why.' Assign cards: Group 1 $\rightarrow -\sqrt{3}$; Group 2 $\rightarrow \pi - 3$; Group 3 $\rightarrow 2.333...$; Group 4 $\rightarrow \sqrt{2}$; remaining cards (-1.5, 0, $2/3$, 3) defended by volunteers. For each defence, prompt the defending group: 'State your decimal approximation. State two numbers it falls between. State your	Mathematical Court Argumentation (SEP: Engaging in Argument from Evidence): learners construct and evaluate claims about number placement using decimal evidence. This mirrors how engineers and scientists justify measurements — you cannot write the exact value, but you can prove your approximation is better than an alternative. Connecting back to the sufuria gap makes the abstraction meaningful: the 'wrong' peg position costs the daughter a small but real piece of string.	Listen for three specific things during defences: (1) Does the learner cite a decimal approximation as evidence (not just 'it looks about right')? (2) Does the learner correctly distinguish 'I can approximate it' from 'I can write it exactly'? (3) For 2.333... — do learners recognise it as rational ($= 7/3$) even though it is non-terminating? Learners who say 2.333... is irrational because 'it goes on forever' need immediate targeted questioning: 'Does it repeat? What fraction equals 2.333...?' This is the most

Academy (English - US curriculum) Search ARES for similar readings	columns for Number Decimal Form Rational or Irrational? Exact position knowable? How the daughter's string problem connects. They then revisit the Ugali Pot Problem: 'The daughter's calculator showed 87.9645... She pegged her string at 88 cm. On our class number line, show me — with your finger — the tiny gap between 88 and the true irrational value $\pi \times 28$.' Teacher leads a whole-class discussion connecting irrational placement to the sufuria string gap.	justification.' WAIT 12 seconds after each defence before opening to the court. Cold-call at least 2 challengers per card from different groups. For irrational placements, push with: 'You said $\sqrt{2}$ is approximately 1.4142. Can you write it exactly? If not, how do you KNOW it belongs exactly there and not one millimetre to the right?' WAIT 15 seconds — this is the key productive struggle moment. After all 8 cards are settled, transition to the phenomenon connection: 'Now think about the daughter and the sufuria. Her calculator said 87.9645... — she cut at 88. On the number line of circumferences, is 88 exactly at the irrational point $\pi \times 28$? What does the tiny gap represent in real life?' Cold-call 3 learners. Validate answers: the gap is real, it matters in engineering and cooking, and it is WHY mathematicians developed approximation techniques. Close: 'For the Kericho temperatures — Analyst groups, which temperature was irrational and how did you know?' Cold-call 2 groups. Confirm that $\sqrt{5}$ above a baseline is irrational because $\sqrt{5}$ is irrational, while 18.75, 19.5, 20, 21 are all rational.		common misconception at this stage and must be addressed before Lesson 5.
Driving Question Board (DQB) Creation  VIDEO: Comparing decimal numbers on a number line Source: Khan Academy (English - US curriculum) Search ARES for similar videos	Learners individually write one sticky note (or card) to add to the class DQB. The prompt has two parts: (A) 'Something I can now answer about the driving question that I couldn't answer in Lesson 3:' and (B) 'A new question the number line activity raised for me:' Examples of expected new questions: 'Can we ever run out of irrational numbers between two rationals?', 'Is there a number that	Distribute sticky notes. Say: 'You have 4 minutes. On your sticky note, write TWO things: first, one thing about our driving question you can now answer better after today's number line activity — and second, one new question today raised for you. Write your name small in the corner.' After 4 minutes, collect notes, read 5–6 aloud (choose a spread of ability levels). As you read each: 'Does	Collaborative Knowledge Mapping on the DQB: clustering learner-generated questions publicly shows the class that mathematical understanding is cumulative and that new knowledge creates new questions rather than closing inquiry. This metacognitive strategy helps learners see their own growth across the 8-lesson sequence.	Read all sticky notes before the next lesson. Count: (1) How many learners correctly state that irrational numbers have exact positions even if unwriteable? (2) How many learners still conflate non-terminating with irrational (mentioning 2.333... as irrational)? (3) What proportion of new questions are about density of irrationals vs. about computation? Use this to calibrate the opening of

 HTML: Square Roots and Irrational Numbers (Flexbook) Source: CK-12  Search ARES for similar readings	is neither rational nor irrational?', 'How did mathematicians first prove π goes on forever?', 'Does every point on the number line have a number, or are there gaps?' The teacher reads out 5–6 contributions aloud, clusters them on the DQB under two headings ('What we now know' and 'What we still wonder'), and explicitly connects the new anchoring question 'Can we always find the exact spot for an irrational number?' to the DQB.	anyone want to add to this thought?' WAIT 10 seconds. Sort notes onto the DQB in real time. Point to the 'What we still wonder' cluster and say: 'Look at how many new questions one number line raised. This is what mathematics feels like — every answer opens three doors. Our driving question asks why we can never write some numbers down completely. Today you proved that those numbers still have a home — a precise spot — on the number line. That is extraordinary.' Point to the sufuria photograph on the wall (if displayed): 'The daughter's string landed at 88. The true spot for $\pi \times 28$ is just to the left of that. Always. No matter how many decimal places her calculator shows.'		Lesson 5.
Model Building Phase  VIDEO: Approximating square roots Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Square Roots and Irrational Numbers Practice Source: CK-12  Search ARES for similar readings	Learners return to their cumulative 'Real Number System Model' (a diagram started in Lesson 1 showing nested sets: Natural $\rightarrow$ Whole $\rightarrow$ Integer $\rightarrow$ Rational $\rightarrow$ Real, with Irrational as the complement of Rational within Real). They now annotate it with three new additions: (1) They draw a number line inset on their model, marking the approximate positions of $\sqrt{2}$, $-\sqrt{3}$, $\pi - 3$, and 2.333... with dotted arrows showing 'irrational' vs. 'rational' landing zones. (2) They add a speech bubble to the 'Irrational' region: 'These numbers have exact positions on the number line even though we can never write them completely.' (3) They add a 'Kenyan Connections' box listing: sufuria circumference ($\pi \times 28$), Kericho temperature anomaly ($\sqrt{5}$ above baseline), alongside known rational examples (unga price Ksh 60.50 =	Distribute learners' individual model sheets (collected at the end of Lesson 3). Say: 'Your model of the real number system needs an upgrade. In the next 8 minutes, add three things — I will write them on the board.' Write the three additions clearly. Circulate and look specifically at the 'Kenyan Connections' box: 'Tell me — why is 60.50 rational? Can you write it as a fraction?' WAIT 12 seconds. 'Why is the sufuria circumference irrational? What about 160 km?' Push learners to articulate: 'It's not about whether it's big or small — it's about whether it equals a ratio of two integers.' After 8 minutes, ask groups to share with neighbours: 'Explain one annotation to your neighbour — they must be able to tell me what you added and why, so make it clear.' Close the lesson: 'Your model now shows that the real	Progressive Model Revision (SEP: Developing and Using Models): learners iteratively refine their nested-set diagram to incorporate the number line dimension. Adding the 'Kenyan Connections' box grounds the abstract model in local reality and reinforces transfer — the skill of classification applies to sufuria measurements and Kericho temperatures alike, not just textbook examples.	Collect models at the end of the lesson. Check three things: (1) Is the number line inset correctly annotated — are irrational numbers placed in the correct region relative to integers? (2) Is the speech bubble mathematically accurate (exact position vs. exact written form)? (3) Does the 'Kenyan Connections' box correctly classify all listed examples? Any learner who classifies 160 km as irrational (because 'it is a measurement') needs a targeted discussion at the start of Lesson 5 distinguishing measurement imprecision from mathematical irrationality.

	121/2, Nairobi–Nakuru distance 160 km = 160/1). Groups briefly share their updated models with a neighbouring group, explaining one new annotation each.	number line has no gaps — every number, rational or irrational, has a home. But can we always find that home precisely? That question goes onto our DQB and drives our next lesson, where we explore operations with irrational numbers.' Collect models for teacher review.		
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record brief notes: (1) PLACEMENT ACCURACY — When groups pegged their cards to the string number line, how accurate were the irrational placements compared to the rational ones? What does this suggest about learners' comfort with decimal approximation? (2) THE 2.333... MISCONCEPTION — Did learners initially classify 2.333... as irrational because it is non-terminating? How many? Did the Mathematical Court discussion resolve this, or do you need to revisit the 'repeating = rational' rule explicitly in Lesson 5? (3) KERICHO TEMPERATURES — Did learners engage authentically with the Kericho data, or did it feel like an add-on? Would a different Kenyan context (e.g. Rift Valley rainfall data, Kisumu lake temperature) have been more resonant for your specific class? (4) PHENOMENON CONNECTION — During the sufuria gap discussion in Phase 3, were learners able to articulate WHY the gap exists (irrational circumference vs. rational cut)? If not, consider beginning Lesson 5 with a 2-minute re-enactment using an actual piece of string and a circular object. (5) DQB QUALITY — What new questions appeared on the DQB that you had not anticipated? Do any of these questions open a productive thread for Lessons 5–8 (operations with irrationals, rationalising denominators, limits of approximation)?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners constructed a classroom string number line (–5 to 5) and pegged 8 number cards (including $-\sqrt{3}$, $\sqrt{2}$, $\pi - 3$, 2.333...) to their correct positions after computing decimal approximations. They also plotted and classified 7 Kericho daily temperatures as rational or irrational on a mini number-line strip.
What did I learn?	Every real number — whether rational or irrational — occupies a unique, precise point on the number line. Irrational numbers like $\sqrt{2}$ and $\pi - 3$ cannot be written exactly as fractions or terminating decimals, but they can be located to any desired accuracy using decimal approximations. Non-terminating repeating decimals (e.g. 2.333...) are rational. The daughter's 88 cm string is a rational approximation pegged near — but never exactly at — the irrational circumference $\pi \times 28$.
How does this explain the phenomenon?	The real number line is complete and continuous: between any two rational numbers there are infinitely many irrational numbers, and every irrational number has a definite address on the line even though we can never write that address in full. This is why the ugali-pot daughter's string always has a tiny gap — she is placing a rational peg at an irrational address, and no matter how many decimal places her calculator shows, she can only get closer, never exact.

LESSON 5 (40 minutes): Operating on Real Numbers — Can We Calculate Exactly with Numbers That Never End?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners apply the four operations and BODMAS to real numbers — including integers, fractions, decimals, and surds — and investigate what happens to rational and irrational numbers when they are combined, in order to advance understanding of why the sufuria's circumference can never be computed exactly.
Knowledge	– The four operations (addition, subtraction, multiplication, division) can be applied to all real numbers. – BODMAS (Brackets, Of/Orders, Division, Multiplication, Addition, Subtraction) governs the order of operations for real numbers. – The sum, difference, or product of two rationals is always rational; combining a rational with an irrational generally produces an irrational; but some combinations of irrationals can be rational (e.g. $\sqrt{2} \times \sqrt{2} = 2$). – Approximation introduces small but unavoidable errors; working with the exact symbolic form (e.g. 28π) preserves precision.
Skills	– Apply BODMAS correctly to multi-step calculations involving integers, fractions, decimals, and irrational numbers. – Decide, with justification, whether the result of combining two real numbers will be rational or irrational. – Choose an appropriate level of approximation for a given real-life context (e.g. cutting string, pricing unga, measuring temperature).
Attitudes	– Appreciate that mathematical precision matters in everyday Kenyan professional contexts — from construction in Nairobi to pharmacy dosage calculations. – Show perseverance and care when working with non-terminating numbers, recognising that keeping the symbolic form is often more accurate than premature rounding.
Key Inquiry Question	How do we use real numbers in day-to-day situations?
Purpose in Storyline	Lessons 1–4 established what real numbers are and where they live on the number line. Lesson 5 now asks: what happens when we operate on them? Learners discover that arithmetic on real numbers follows consistent rules (BODMAS), but that the rational/irrational character of results depends on which types of numbers are combined — directly explaining why $28 \times \pi$ cannot be 'simplified away' into a nice fraction and why the daughter will always face a gap or overlap when she cuts her string.
Safety Notes	No specific hazards. If physical string and a sufuria are used as props, ensure the sufuria rim has no sharp burrs.

B. LESSON OVERVIEW

This lesson is the operational heart of the Real Numbers sequence. Having classified, represented, and located real numbers on the number line in Lessons 1–4, learners now compute with them. The lesson opens by returning to the anchoring phenomenon: the daughter calculated $28 \times \pi$ on her calculator and got 87.96459430.... She rounded to 88 cm and cut her string — but there was always a tiny gap or overlap. The central question driving this lesson is: does it matter in what order we do calculations with real numbers, and what kind of number do we get when we mix rationals and irrationals?

Learners begin with a prediction task — estimating, without a calculator, whether combining specific pairs of Kenyan-context numbers (e.g. the price of unga plus the cost of charcoal; the circumference of the sufuria minus a rational approximation) will yield a rational or irrational result. They then work through a structured guided-inquiry activity using real Kenyan marketplace and construction contexts to discover the closure and combination rules for real number operations, and to apply BODMAS in multi-step calculations that include surds and π .

The lesson closes with learners revising the class model of the real number system to annotate it with operation results: arrows showing 'rational + rational = rational', 'rational $\times$ irrational = irrational (usually)', and 'irrational $\times$ irrational = sometimes rational'. This cumulative model directly feeds the next lesson on indices and standard form, and keeps the sufuria phenomenon alive as the running thread.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Adding... Subtracting... Multiplying and Dividing - Missing Operation - Order of Operations Source: CK-12  Search ARES for similar videos  EXERCISE: Operations with Real Numbers Practice Source: CK-12  Search ARES for similar readings	Each learner receives a half-sheet with six 'Number Pairs' drawn from Kenyan contexts (see below). Without using a calculator, they write R (rational) or I (irrational) to predict whether the result of the stated operation will be rational or irrational, then share their prediction with a partner using a Think-Pair-Share. The six pairs are: (1) $60.50 + 34.5$ [price of unga + temperature in Kisumu in $^{\circ}\text{C}$]; (2) $\pi + 3$; (3) $\sqrt{2} \times \sqrt{2}$; (4) $28 \times \pi$ [the sufuria circumference]; (5) $(1/3) \div (2/3)$; (6) $\sqrt{3} + \sqrt{5}$. Learners record their reasoning in one sentence per pair.	Distribute the half-sheets face-down. Say: 'Before we calculate anything today, I want your gut feeling. You have 3 minutes — silent, individual thinking first. Flip your sheet.' [Allow 3 minutes of silent work.] After 3 minutes: 'Now turn to your partner — 90 seconds each to compare answers. If you disagree, write down WHY you disagree.' [Allow 3 minutes pair discussion.] Cold-call 4 learners (aim for mixed confidence levels): 'Amina, what did you predict for pair 4 — the sufuria one — and why?' [WAIT TIME: 12 seconds after posing each question before accepting a response.] After each response, ask the class: 'Hands up if you agreed with Amina. Hands up if you disagreed.' Do NOT confirm or correct answers yet. Write the six predictions on the board in a T-chart: PREDICT: Rational PREDICT: Irrational. Say: 'We are going to test every one of these predictions today. Keep your sheet — you'll revise it at the end.'	Activate-and-Surface (Elicitation): By surfacing predictions before instruction, the teacher makes prior conceptions visible. Disagreements — especially on pairs 3 ($\sqrt{2} \times \sqrt{2} = 2$, which is rational and surprises most learners) and 4 ($28 \times \pi$, irrational) — create productive cognitive conflict that the Observe Phase will resolve. Recording predictions publicly on a T-chart creates accountability and makes later revision meaningful.	Teacher scans the room during silent work: are learners writing reasoning or just guessing R/I? Target indicator: learner can articulate at least one reason (e.g. 'both are fractions so it stays a fraction'). Listen during cold-calls for misconceptions: if learners claim $\sqrt{2} \times \sqrt{2}$ is irrational, note this for targeted instruction in the Explain Phase. If more than half the class predicts pair 4 is rational, flag this as the central misconception to address.
Observe Phase  VIDEO: Selecting the First Operation - Order of Operations	Learners work in groups of 4 on a structured 'Calculation Station' worksheet with three sections. Section A (BODMAS warm-up, 3 questions): e.g. 'A mama mboga in Gikomba charges Ksh 60.50 for a 2 kg packet of unga and Ksh 15.00	Distribute worksheets. Say: 'Section A is a BODMAS workout — show every step, no steps skipped. Section B is the investigation: write both the exact form AND the decimal. Section C is your detective work — what	Structured Inquiry with Evidence Recording: The three-section worksheet scaffolds learners from familiar BODMAS practice (Section A) into novel territory (Section B) and then into pattern recognition (Section C). Writing	Key observable: in Section B, can learners correctly identify $\sqrt{2} \times \sqrt{2} = 2$ (rational)? This is the highest-leverage misconception. If 3 or more groups get it wrong, plan a whole-class pause during the Explain Phase. In Section C, look

Source: CK-12 Search ARES for similar videos HTML: Connect Variable Expressions and the Order of Operations with Real-World Problems (Flexbook) Source: CK-12 Search ARES for similar readings	for 500 g of sukuma wiki. She also gives a discount of Ksh 5.25. Calculate the total cost without a calculator, showing all steps in BODMAS order.' Section B (Combining number types, 4 questions): learners compute (i) $1/4 + 0.75$, (ii) $\pi + 3$, (iii) $\sqrt{2} \times \sqrt{2}$, (iv) $28 \times \pi$ — writing the exact symbolic answer AND a decimal approximation to 4 d.p. for each. Section C (Pattern hunt, 1 question): 'Look at your four answers in Section B. Make a conjecture: what determines whether your answer is rational or irrational?' Groups record their conjecture as a full sentence.	pattern do you see?' [Allow 12 minutes of group work.] Circulate actively, targeting these check-ins: (1) At every group, look at Section B part (iii): if they wrote $\sqrt{2} \times \sqrt{2} = \sqrt{4} = 2$ correctly, ask 'Is 2 rational or irrational?' [WAIT TIME: 10 seconds]. (2) For Section B part (iv), ensure groups write '28π' as the exact form — not just the decimal. If a group writes only '87.9646', say: 'You have the approximation — brilliant. But what is the EXACT answer? How would a mathematician write it?' [WAIT TIME: 12 seconds.] (3) At the halfway point, cold-call one group: 'Kamau's group — read me your conjecture from Section C so far.' After they respond, ask the class: 'Does anyone have a different conjecture?' Cold-call 2 more groups. Do not validate yet. At the end of group work, ask all groups to hold up their Section C conjecture sentence for a gallery walk (30-second silent read of neighbours' conjectures).	both exact and decimal forms side-by-side in Section B forces learners to confront the difference between an approximation and an exact answer — directly mirroring the daughter's situation with 28π vs 87.96 cm. The gallery walk surfaces diverse conjectures without premature teacher closure.	for conjectures that mention 'type of number' or 'whether π or $\sqrt{}$ is involved' — these show readiness for the formal rule. Conjectures that only mention 'size of numbers' indicate the learner is not yet thinking about number classification.
Explain Phase VIDEO: Determinant after row operations Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Operations with Real Numbers Practice Source: CK-12 Search ARES for similar readings	The teacher leads a whole-class debrief structured around four 'result cards' posted on the board (one per Section B question). For each card, a group representative states their answer; the class votes; the teacher confirms and annotates. Learners then receive a short reference note (one side of A5) titled 'Rules for Combining Real Numbers' with four rules stated in plain language plus one symbolic example each. Learners immediately apply these rules to three new 'Nairobi context' problems: (1) A jua kali fabricator in Kamukunji needs to cut a steel rod of length $(3 + \sqrt{5})$ metres into two equal pieces — is each piece	Post the four result cards. For each: 'Group 2 — Wanjiru, read your answer for question (iii).' [WAIT TIME: 10 seconds.] 'Class — thumbs up if you agree, thumbs sideways if you're not sure, thumbs down if you disagree.' Count responses visibly. For $\sqrt{2} \times \sqrt{2} = 2$: say 'This one surprises a lot of people. Let's see why it's rational.' Write on the board: $\sqrt{2} \times \sqrt{2} = (\sqrt{2})^2 = 2$. Say: 'Two irrational numbers can multiply to give a rational number. This is NOT what most of us predicted.' Pause for effect. Distribute the reference note. Read through the four rules aloud together. Then say: 'Three application problems — I want	Claim-Evidence-Reasoning (CER) with Counter-Example Emphasis: Each application problem is framed as a claim to justify (rational or irrational?) with evidence (the calculation) and reasoning (which rule applies). The $\sqrt{2} \times \sqrt{2}$ counter-example is treated as a 'surprise result' deliberately — research shows that counter-intuitive cases deepen schema revision. The problem about 22/7 explicitly reconnects to the phenomenon and surfaces the precision-vs-exactness trade-off that is the driving question's core tension.	For problem 3, the target indicator is: learner can state both that $28 \times 22/7 = 88$ (rational) AND that $22/7 \neq \pi$, therefore the answer is an approximation, not exact. If learners say only '88, rational' without noting the approximation issue, probe: 'Is 22/7 the same as π?' This reveals whether the learner has integrated the lesson's core idea. Collect half-sheets at end of Explain Phase — scan for correct application of all four rules.

	rational or irrational? (2) A pharmacist in Kericho measures two doses: 0.333... litres and $\frac{1}{6}$ of a litre of a syrup — what is the total, and is it rational? (3) The daughter recalculates the sufuria rim: she uses $\frac{22}{7}$ instead of π — is $28 \times \frac{22}{7}$ rational or irrational, and why does this matter for her string? Learners write answers individually, then compare with a partner.	SILENT individual work for 4 minutes first, then you compare with your partner. Go.' [4 minutes silent, then 2 minutes pair comparison.] Cold-call on problem 3: 'Odhiambo — if the daughter uses $\frac{22}{7}$ instead of π, is $28 \times \frac{22}{7}$ rational or irrational?' [WAIT TIME: 12 seconds.] After the answer, say: 'So by switching to $\frac{22}{7}$, she made the calculation rational — she got exactly 88 cm. But wait — is $\frac{22}{7}$ actually equal to π?' [WAIT TIME: 10 seconds.] 'No — it is only an approximation. So she introduced an error in exchange for being able to write the answer down. That is the trade-off at the heart of our driving question.'		
Driving Question Board (DQB) Creation  VIDEO: Adding, Subtracting, and Multiplying - Missing Number - Order of Operations Source: CK-12  Search ARES for similar videos  EXERCISE: Operations with Real Numbers Practice Source: CK-12  Search ARES for similar readings	Each group receives two sticky notes of different colours: one ORANGE ('Something we can now answer about the driving question') and one BLUE ('A new question this lesson raised for us'). Groups discuss for 90 seconds, then a representative posts both notes on the class DQB under the driving question banner: 'Why can we never write some everyday numbers down completely — and how do mathematicians and Kenyan professionals work accurately with numbers that go on forever?' The class does a 2-minute silent gallery walk reading all new notes. The teacher then highlights two orange notes and two blue notes aloud, connecting them explicitly to the lesson's findings.	Hand out sticky notes. Say: 'Orange note: what can we NOW answer about our big question that we couldn't before today? Blue note: what new question did today's lesson raise for you? You have 90 seconds as a group.' [90 seconds group discussion.] 'One representative — post both notes on the DQB. Everyone else: stand up and do a silent gallery walk for 2 minutes. Read what other groups wrote.' [2 minutes silent gallery walk.] After gallery walk, teacher reads two orange notes aloud: e.g. 'Group 3 says: we can now answer WHY the daughter always gets a gap — because $28 \times \pi$ is irrational and can never be written exactly, no matter how many decimal places.' Say: 'Exactly. That is a real answer to our driving question.' Then read a blue note: e.g. 'Group 1 asks: can computers store π exactly?' Say: 'That is a beautiful new question. Does anyone want to take a guess	Two-Colour DQB Protocol: Using two distinct note colours forces learners to distinguish between resolved understanding and emergent questions, preventing the DQB from becoming a mere display. The gallery walk ensures every learner reads peers' thinking, not just their own group's. The teacher's act of reading and validating specific notes models metacognitive reflection — showing that both 'answers' and 'new questions' are valued outcomes of scientific and mathematical inquiry.	Quality indicator for orange notes: does the note reference the phenomenon (the sufuria, the daughter, 28π) or use correct mathematical vocabulary (rational, irrational, approximation)? Vague notes like 'we learned about BODMAS' indicate surface processing. Quality indicator for blue notes: does the question push forward into territory not yet covered (e.g. indices, surds, infinite series) or reveal a residual misconception? Collect blue notes at the end — they directly inform Lesson 6 planning.

		before next lesson?' [WAIT TIME: 12 seconds. Cold-call 2 volunteers.] Close by pointing to the DQB: 'Look at how much ground we have covered since Lesson 1. We predicted, we classified, we placed numbers on the line, and now we operate on them. Each lesson, our model gets richer.'		
Model Building Phase  VIDEO: Adding and Subtracting with Parentheses - Missing Number - Order of Operations Source: CK-12  Search ARES for similar videos  EXERCISE: Operations with Real Numbers Practice Source: CK-12  Search ARES for similar readings	Learners retrieve the cumulative class model of the real number system — a nested-set diagram ($\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$, with $\mathbb{R} \setminus \mathbb{Q}$ labelled as irrationals) built across Lessons 1–4. Working individually in their exercise books, each learner adds an 'Operations Layer' to their personal copy of the model: four annotated arrows showing what type of number results when each operation is performed between: (i) rational + rational, (ii) rational + irrational, (iii) irrational + irrational, (iv) irrational $\times$ irrational. They label the sufuria calculation ($28 \times \pi$) on their model with an arrow from 28 (in $\mathbb{Q}$) and π (in $\mathbb{R} \setminus \mathbb{Q}$) — annotated: 'Exact answer — irrational. Approximate answer 88 cm — rational.' They add one sentence at the bottom: 'Today's lesson changes my understanding of the driving question because...'	Say: 'Get out your personal model from your exercise book — the nested diagram we have been building since Lesson 1. Today we add the Operations Layer.' Display the class model on the board. 'I want four arrows on your diagram. Each arrow starts at two numbers — one on each side — and points to what type of number the result is. Use the rules from your reference note.' [Allow 4 minutes of individual model revision.] Circulate. If a learner is only writing labels without arrows, say: 'Show me the journey — where do 28 and π live in your diagram, and where does 28π land?' [WAIT TIME: 10 seconds.] After 4 minutes: 'Now write your one sentence at the bottom. Finish the stem: Today's lesson changes my understanding of the driving question because...' [Allow 90 seconds.] Cold-call 3 learners to read their sentences. [WAIT TIME: 10 seconds per learner.] After each, ask the class: 'Does anyone want to add to what Njeri said?' Close: 'Your model now shows not just the number families, but how they behave when you combine them. Next lesson, we will zoom in on a special family of real numbers — those written as powers — and see what new tools that gives us for the sufuria problem and	Cumulative Model Revision (Iterative Modelling): The model is not created fresh each lesson but deliberately built up over the sequence, making the growth of understanding visible and tangible. Adding the 'Operations Layer' as arrows (rather than static labels) foregrounds the dynamic, relational nature of number classification — numbers don't just exist in categories, they interact across categories. The sentence stem 'Today's lesson changes my understanding because...' is a metacognitive forcing function that consolidates lesson-level learning into the larger storyline arc.	Target: learner's model correctly shows that rational $\times$ irrational $\rightarrow$ irrational (with $28 \times \pi$ as the example) AND that irrational $\times$ irrational can be rational (with $\sqrt{2} \times \sqrt{2} = 2$ as the example). If both are present, the learner has grasped the lesson's conceptual core. Sentence stems are collected or photographed as an exit indicator — look for explicit reference to the driving question tension (exact vs approximate) rather than just restating 'we learned BODMAS'. Students who write only procedural sentences (e.g. 'I now know how to do BODMAS') should be targeted for a brief check-in at the start of Lesson 6.

		beyond.'		
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D. TEACHER REFLECTION

1. During the Predict Phase, which of the six number pairs generated the most disagreement, and what does that tell you about your learners' residual misconceptions about rational vs irrational numbers going into Lesson 6?
2. Were learners able to write ' 28π ' as an exact symbolic answer, or did most groups jump straight to the decimal approximation? If the latter, what instructional move could you use in Lesson 6 to reinforce the value of exact symbolic forms?
3. The application problem about $22/7$ vs π is designed to surface the precision trade-off at the heart of the driving question. Did learners articulate both parts of the answer ($28 \times 22/7 = 88$ is rational AND $22/7 \neq \pi$, so it is an approximation)? If not, how will you revisit this in the next lesson?
4. Looking at the blue sticky notes from the DQB, what new questions did learners raise? Do any of these questions connect to upcoming content in the Real Numbers sub-strand (e.g. indices, surds, standard form), and can you use them as an entry point for Lesson 6?
5. How rich were the 'Operations Layer' additions to learners' personal models? Did the models show arrows and relational thinking, or just static labels? What scaffolding might help weaker learners make their models more dynamic before the next lesson?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners computed the results of adding, subtracting, multiplying, and dividing pairs of real numbers drawn from Kenyan marketplace and construction contexts (unga prices, the sufuria circumference, pharmacist doses in Kericho, jua kali fabrication in Kamukunji). They observed that most combinations follow predictable rules — but that irrational $\times$ irrational can sometimes yield a rational result ($\sqrt{2} \times \sqrt{2} = 2$), and that replacing π with $22/7$ makes the sufuria calculation rational but introduces an approximation error.
What did I learn?	The four operations on real numbers obey BODMAS. When two rationals are combined by any operation (result defined), the result is always rational. When a rational is combined with an irrational by addition or subtraction, the result is always irrational. When a rational is multiplied or divided by an irrational, the result is generally irrational. When two irrationals are combined, the result can be either rational or irrational — there is no single rule. Keeping the exact symbolic form (e.g. 28π) is more precise than any decimal approximation.
How does this explain the phenomenon?	The daughter's calculator shows 87.96459430... because $28 \times \pi$ is the product of a rational (28) and an irrational (π), making the result irrational — a number that cannot be written as a terminating or repeating decimal, no matter how many digits the calculator displays. When she rounds to 88 cm or uses $22/7$ instead of π , she converts the answer into a rational number — gaining the ability to write it down exactly — but at the cost of introducing a small error. This is why there is always a tiny gap or overlap in her string: exactness and writability are fundamentally in tension for irrational numbers.

LESSON 6 (80 minutes): Operating on Real Numbers: BODMAS in the Market and the Sufuria

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To enable learners to apply the four operations (addition, subtraction, multiplication, division) to real numbers — including integers, fractions, decimals, and irrational numbers — using BODMAS/BODMAS rules in multi-step expressions set in Kenyan market and measurement contexts.
Knowledge	Learners understand that the order of operations (BODMAS) governs the evaluation of any multi-step expression involving real numbers, whether the numbers are rational (e.g., prices in shillings, weights in kg) or irrational (e.g., π used in circumference calculations). They know that approximating an irrational number before or after applying operations can yield different levels of accuracy.
Skills	Learners can: (1) correctly apply BODMAS to multi-step expressions involving mixed numbers, decimals, fractions and irrational numbers; (2) choose an appropriate degree of approximation for a given real-life context; (3) identify and correct common operational errors through peer-marking.
Attitudes	Learners appreciate that mathematical accuracy and systematic use of operation rules are essential professional skills — from a mama mboga calculating market costs to an engineer computing dimensions of a sufuria rim — and develop confidence in tackling complex multi-step problems.
Key Inquiry Question	How do we use real numbers in day-to-day situations?
Purpose in Storyline	In previous lessons learners classified real numbers and explored rational and irrational numbers through the sufuria circumference puzzle. Now they must actually compute with those numbers: market transactions involve mixed rational quantities (kg of unga at Ksh 60.50 per kg, bundles of sukuma wiki at Ksh 15, tomatoes at Ksh 5 each) and the circumference formula reintroduces π . This lesson answers: once we know what kind of number we have, how do we operate on it correctly and in the right order?
Safety Notes	No physical hazards. Ensure calculators or printed mathematical tables are available for every pair. Remind learners that calculator displays are truncated approximations of irrational numbers — a conceptual 'safety note' to prevent over-reliance on calculator answers without reasoning.

B. LESSON OVERVIEW

Lesson 6 of 8 in the evidence-gathering sequence. Learners move from classifying real numbers (Lessons 1–5) to operating on them. The lesson opens with a Predict phase in which learners estimate a multi-step market bill before computing it formally. The Observe phase presents three worked Kenyan market scenarios (ugali flour + sukuma wiki + tomatoes) structured as BODMAS expressions of increasing complexity, which pairs solve and then peer-mark using teacher-issued answer keys. The Explain phase surfaces the two most common errors — ignoring brackets and premature rounding of π — through whole-class discussion, with the sufuria circumference formula as the unifying thread. The DQB is updated to record the insight that "operation rules apply equally to rational and irrational numbers, but rounding irrational numbers mid-calculation amplifies error." The lesson closes with a Model-Building phase in which learners revise their cumulative real-number models to show how the four operations interact with the rational/irrational distinction.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Order of operations example: fractions and exponents Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Equations with square roots: decimals & fractions Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners individually read the 'Gikomba Morning Bill' prompt on their activity cards: 'Mama Wanjiku buys $2\frac{1}{2}$ kg of unga at Ksh 60.50 per kg, 3 bundles of sukuma wiki at Ksh 15 each, and 8 tomatoes at Ksh 5 each. She pays with a Ksh 500 note. Without using a calculator, predict: (a) the total cost, and (b) the change she receives. Write down the single expression you would evaluate, then estimate the answer.' Learners record predictions in their notebooks and share with a shoulder partner for 2 minutes.	Distribute activity cards. Read the prompt aloud once, then say: 'Before you calculate anything, write the expression first — show me the maths sentence, not just a number.' Allow 4 minutes of silent individual work. After 2 minutes of partner sharing, cold-call 3 learners to read their expressions aloud (select one who used a fraction, one who used decimals, one who wrote it incorrectly). For each, ask: 'Does everyone agree this expression matches the story? Why or why not?' Apply 12 seconds WAIT TIME after each question. Do NOT confirm or correct answers yet — record expressions on the board under the heading 'Our Predictions'.	Estimation before calculation (Predict-Before-Compute). By committing to an expression and an estimate, learners activate prior knowledge of BODMAS structure and expose implicit assumptions about operation order. This creates productive cognitive dissonance when the computed answer differs from intuition.	Scan notebooks for correct expression structure: look for brackets around the multiplication sub-expressions, e.g., $(2.5 \times 60.50) + (3 \times 15) + (8 \times 5)$. Flag learners who wrote a running sum without brackets — these learners need targeted support in the Observe phase. Observable indicator: at least 70% of learners write a structurally correct multi-term expression before the Observe phase begins.
Observe Phase  VIDEO: Evaluating expressions with variables: exponents Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Order of operations with fractions and exponents Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners work in assigned pairs. Each pair receives a worksheet titled 'Gikomba Market: Three BODMAS Challenges'. The three problems escalate in complexity: (Level 1) Evaluate $(2\frac{1}{2} \times 60.50) + (3 \times 15) + (8 \times 5)$ — total market bill; (Level 2) Mama Wanjiku also buys a piece of string to go around her sufuria (diameter 28 cm). String is sold per centimetre at Ksh 0.50. Using $\pi \approx 3.142$, evaluate $0.50 \times (\pi \times 28)$ and compare with $0.50 \times (3.142 \times 28)$ — discuss any difference; (Level 3) A market stall charges Ksh $[4^2 \div (\sqrt{16} - 1) + 2\frac{1}{2}]$ per 100 g of nyama choma. Evaluate the expression and find the cost of 350 g. After 15 minutes of pair work, pairs exchange worksheets with an adjacent pair and mark using a teacher-issued worked solutions sheet. Pairs record marks and circle any	Circulate during pair work. Stop at pairs who are working on Level 2 and ask: 'What happens to your answer if you round π to 3.142 before multiplying, versus rounding after? Try it both ways.' Use 12 seconds WAIT TIME. At 15 minutes, pause the class: 'Pens down. Pass your worksheet one pair to your right.' Distribute worked solution sheets face-down and say: 'Do not flip until I say go.' Check that every pair has a worksheet to mark. Say 'Go.' After 5 minutes of peer-marking, cold-call 2 pairs to report a discrepancy they found, asking: 'What was the error? Which step of BODMAS was violated?' Record the two most common errors on the board.	Peer-Marking with Worked Solutions (Structured Error Analysis). Having learners mark each other's work against a fully worked solution — rather than just a final answer — forces them to trace errors to a specific operational step. This mirrors the professional practice of a Kenyan accountant or quantity surveyor checking a bill of quantities line by line.	Collect marked worksheets at the end of the phase. Scan for: (1) Level 1 — correct total Ksh 242.25, change Ksh 257.75; (2) Level 2 — correct string cost Ksh $43.988 \approx \text{Ksh } 44.00$ using $\pi \approx 3.142$ (learners should note this is still an approximation because π is irrational); (3) Level 3 — correct step-by-step: $4^2 = 16$, $\sqrt{16} = 4$, $4 - 1 = 3$, $16 \div 3 = 5.333\dots$, $5.333\dots + 2.5 = 7.833\dots$, cost per 100g $\approx \text{Ksh } 7.83$, cost of 350g $\approx \text{Ksh } 27.42$. Observable indicator: fewer than 3 pairs make a BODMAS ordering error on Level 1 after peer-marking discussion.

	discrepancies in red pen.			
Explain Phase  VIDEO: Order of operations example Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Order rational numbers Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Whole-class discussion facilitated by the teacher. Learners revisit the two errors flagged on the board. For each error, the class works through the correct solution step by step, with individual learners called on to supply the next step. Learners then respond in writing (2 minutes) to the prompt: 'The daughter in the Gikomba story approximated π as 3.142 before multiplying by 28. Explain in one sentence why this introduces a small error, and state whether BODMAS still applies when one of the numbers is irrational like π.' Learners share responses with their partner, then one response is shared whole-class.	Point to the first error on the board (e.g., addition performed before multiplication). Say: 'Who can tell me which letter of BODMAS tells us what to do first here? Put your hand up if you know — I will pick someone in 12 seconds.' Apply 12-second WAIT TIME. Cold-call 1 learner. If incorrect, say: 'Interesting — does anyone want to build on that?' Cold-call a second learner. Work through the correction live on the board, narrating each BODMAS step. Then transition to the π error: 'Now look at the Level 2 problem. Kioni rounded π before multiplying. Adhiambo rounded after multiplying. Who got a more accurate answer? By how much?' Give 12 seconds WAIT TIME, then cold-call 2 learners. Write the comparison on the board: $(3.142 \times 28) \times 0.50$ vs. $(\pi \times 28) \times 0.50 \approx (87.9646... \times 0.50)$. Say: 'The error from early rounding here is small — but for a Kenyan civil engineer calculating the circumference of a water tank in Kisumu, would that matter?' Allow 10 seconds, then cold-call 1 learner. Close by saying: 'Write your answer to the phenomenon prompt now — 2 minutes, silent.'	Live Error Correction with Think-Aloud (Metacognitive Modelling). The teacher models step-by-step reasoning aloud while learners supply each step, making the invisible thinking process visible. Connecting the rounding error to a professional Kenyan context (civil engineer, water tank) anchors the abstract concept of irrational number precision in a meaningful real-world consequence.	Collect the written one-sentence responses. Look for: (1) acknowledgement that π has infinitely many non-repeating decimal places so any written value is an approximation; (2) confirmation that BODMAS applies regardless of whether numbers are rational or irrational. Observable indicator: at least 80% of written responses correctly state that BODMAS governs operation order for all real numbers, and that rounding early reduces precision.
Driving Question Board (DQB) Creation  VIDEO: Worked example: Order of operations (PEMDAS) Source: Khan Academy (English - US curriculum)	Learners receive a sticky note each (or write in their DQB section of their notebook on the ARES platform). They complete two sentence starters: (1) 'Something we can now answer about the driving question: ...' and (2) 'A new question this lesson raised for me: ...' Responses are shared with the class and the teacher adds the key insight to the class DQB:	Display the driving question on the board: 'Why can we never write some everyday numbers down completely — and how do mathematicians and Kenyan professionals work accurately with numbers that go on forever?' Say: 'We have now worked with these numbers. What can we say today that we could NOT say at the start of Lesson 1? Write your sticky	Driving Question Board Curation (Progressive Sense-Making Record). The DQB functions as a living document of the class's growing understanding. By explicitly linking today's BODMAS insights back to the original phenomenon (the daughter's string problem), learners see how operational fluency with real numbers is one necessary piece of	Review sticky notes / DQB entries for conceptual accuracy. Observable indicator: at least 3 learners connect BODMAS specifically to the irrational number π and the sufuria circumference. Flag any learner who writes 'BODMAS only works for whole numbers' — this misconception must be addressed in Lesson 7.

 Search ARES for similar videos  EXERCISE: Multiply & divide rational expressions: Error analysis Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	'BODMAS applies to ALL real numbers — rational and irrational. Rounding an irrational number early in a calculation reduces accuracy. The daughter in Gikomba could reduce her error by computing $\pi \times 28$ first, then rounding only at the final step.'	note — 3 minutes.' After 3 minutes, cold-call 3 learners to read their 'something we can now answer' note. For each, ask: 'Does this answer part of our driving question? Which part?' After posting insights to the DQB, point to any remaining unanswered DQB questions from earlier lessons and ask: 'Did today's lesson answer any of these? Move it to the answered column if so.'	the full answer to the driving question.	
Model Building Phase  VIDEO: Limits at infinity of quotients with square roots (even power) Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Reduce rational expressions to lowest terms: Error analysis Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners individually revise their cumulative real-number models (begun in Lesson 1 and updated each lesson). Today they must add: (1) a new branch or annotation showing that all four operations apply across the real number system (rational AND irrational); (2) an example expression from today's market context showing correct BODMAS application; (3) a notation warning: 'Irrational numbers (e.g., π, $\sqrt{2}$) — round only at the FINAL step for greatest accuracy.' Learners who finish early add a second example using $\sqrt{16}$ from Level 3 of the worksheet, tracing how it simplified to a rational number mid-calculation.	Say: 'Open your real-number models. Today you discovered something new about how we operate on real numbers. Update your model to show this — you have 12 minutes.' Circulate and look for models. Prompt struggling learners: 'Where did you write rational and irrational numbers? Can you draw an arrow from both to a box labelled BODMAS to show that the same rules apply?' After 12 minutes, ask 2 learners to share their updated model under the visualiser or by holding it up. For each, say to the class: 'One thing you notice, one thing you would add — discuss with your partner for 30 seconds.' Close with: 'Next lesson we will explore a special type of real number — how indices and surds let us write irrational numbers exactly, without approximating at all. How does that connect to our driving question?'	Cumulative Model Revision (Iterative Representation Building). Each lesson's model update requires learners to integrate new knowledge with existing understanding rather than treating concepts as isolated. Adding the BODMAS-over-all-reals annotation forces learners to reconcile their prior (possibly incomplete) conception that operation rules only govern integers or simple fractions.	Collect or photograph updated models. Look for: (1) explicit labelling showing BODMAS applies to both rational and irrational branches of the real number system; (2) at least one correctly structured multi-step expression from today's market context; (3) the 'round at final step' notation. Observable indicator: 75% of models show a structural connection between the rational/irrational classification and the operation rules learned today.

D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) Did the three-level escalation on the worksheet allow all learners to experience success at Level 1 while being sufficiently challenged at Levels 2 and 3? If most pairs stalled at Level 2, consider scaffolding the π approximation comparison with a partially completed table in Lesson 7. (2) Were the two most common errors on the board truly the most diagnostically useful? Review the peer-marked worksheets tonight — if a third error pattern (e.g., treating mixed numbers incorrectly) appears in more than 25% of papers, open Lesson 7 with a targeted 5-minute correction. (3) Did the DQB update generate genuine new questions,

or did learners simply restate what was taught? If the DQB stalled, try a 'What still puzzles me about the sufuria problem?' prompt at the start of Lesson 7. (4) Note which learners have not yet updated their cumulative models in a substantive way across all 6 lessons — these learners may need a structured model template in Lessons 7–8.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners evaluated multi-step BODMAS expressions using real numbers drawn from Gikomba Market scenarios: total market bills involving mixed numbers and decimals (unga, sukuma wiki, tomatoes), string-cost calculations using $\pi \approx 3.142$, and a composite expression involving indices and surds (4^2 , $\sqrt{16}$) applied to nyama choma pricing.
What did I learn?	The order of operations (BODMAS) governs the evaluation of expressions involving ALL real numbers — both rational (prices, weights, distances) and irrational (π , $\sqrt{2}$). Rounding an irrational number before completing all operations introduces avoidable approximation error; the most accurate strategy is to carry the irrational value symbolically (or to maximum decimal places) and round only at the final step.
How does this explain the phenomenon?	When the daughter in Gikomba multiplies $28 \times \pi$ and rounds π to 3.142 first, she applies BODMAS correctly but introduces a small rounding error because π is irrational and 3.142 is only an approximation. BODMAS does not change based on the type of real number — the same rules that govern $\text{Ksh } 60.50 \times 2.5$ also govern $\pi \times 28$. Professional accuracy (e.g., a civil engineer sizing a water-tank rim in Kisumu) requires carrying irrational values to as many decimal places as the context demands and rounding only at the final answer.

LESSON 7 (40 minutes): Absolute Value in Action — How Far From Zero on the Number Line?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners investigate absolute value as the distance of any real number from zero on the number line, applying this concept to both rational and irrational quantities — including the circumference error in the Ugali Pot Problem — to understand how mathematicians and Kenyan professionals measure 'how far off' an approximation really is.
Knowledge	– Absolute value x is defined as the distance of a real number x from zero on the number line, and is always non-negative. – For any real number x: $x = x$ if $x \geq 0$, and $x = -x$ if $x < 0$. – Absolute value applies to ALL real numbers — integers, rationals, and irrationals (e.g. $\pi - 3 = \pi - 3$; $- \sqrt{2} = \sqrt{2}$). – The absolute difference $a - b$ measures the gap between two real numbers, including the error between an exact irrational value and its rational approximation.
Skills	– Evaluate x for integers, fractions, and irrational expressions without a calculator, using the definition. – Apply absolute value to compute approximation error between an exact irrational quantity and a rounded rational value in a real-world Kenyan context.
Attitudes	– Appreciate that 'distance' is always positive, reflecting the mathematical value of precision and the importance of knowing how large an error actually is. – Develop confidence in working with irrational numbers by recognising that absolute value gives a manageable, non-negative measure of any real number's magnitude.
Key Inquiry Question	How do we use real numbers in day-to-day situations?
Purpose in Storyline	Having established (Lessons 1–6) that irrational numbers exist, cannot be written exactly, and must be operated on carefully, learners now gain a powerful tool — absolute value — for quantifying 'how far off' any approximation is. This directly addresses the daughter's string-cutting error: $ 88 - 28\pi $ tells us precisely how large the gap or overlap is, connecting all prior classification and operations work to a concrete, measurable outcome.
Safety Notes	No specific hazards. Ensure learners handle rulers and string samples carefully if physical materials are used.

B. LESSON OVERVIEW

Lesson 7 is the seventh evidence-gathering lesson in the Real Numbers sequence and serves as the crucial bridge between the operations work of Lessons 5 and 6 and the culminating synthesis of Lesson 8. The anchoring phenomenon — the mama mboga's daughter cutting an 88 cm string to wrap the sufuria rim — returns in full focus: learners are asked to compute, precisely, how large the error in that string actually is. This requires the concept of absolute value, which learners explore first through physical intuition (distances along a number line drawn on the classroom floor) and then formally through its definition and algebraic notation.

The lesson moves from concrete to abstract. In the Predict Phase, learners estimate whether the string gap is closer to 0 cm, 0.04 cm, or 4 cm — activating prior BODMAS and approximation knowledge. In the Observe Phase, learners gather evidence by evaluating absolute values across a carefully designed table of real numbers, including negative integers, negative fractions, and irrational expressions such as $|\pi - 3|$ and $|- \sqrt{2}|$. The Explain Phase formalises the piecewise definition of $|x|$ and connects it to the sufuria error calculation $|88 - 28\pi|$, computed using the best decimal approximation of π available. Learners discover that the string error is approximately 0.035 cm — small but non-zero and, crucially, irrational in exact form.

Throughout, Kenyan professional contexts reinforce relevance: a civil engineer in Thika measuring construction tolerances, a pharmacist in Kisumu checking drug dosage errors, and a maize trader in Eldoret computing mass discrepancies all use absolute difference as a daily tool. By the end of the lesson, learners can evaluate absolute values of any real number, compute approximation errors, and articulate — with mathematical precision — why the daughter's string will always have a tiny gap or overlap that is itself an irrational number.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Meaning of absolute value Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Ordering negative number expressions Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners individually read a short scenario on the board: 'The daughter cuts the string to 88 cm. The exact circumference is 28π cm. Is the error closer to: (A) exactly 0 cm, (B) about 0.035 cm, or (C) about 4 cm?' Each learner writes their choice and a one-sentence justification in their exercise books before any discussion. They also sketch a number line fragment and try to mark both 28π and 88 on it, estimating which is larger and by how much.	Display the three options on the board using large writing. Say: 'Before we calculate anything today, I want you to use everything you already know about π and approximation from our last six lessons. Write your prediction and your reason — one sentence. You have 90 seconds. Go.' [Allow 90 seconds of silent individual work.] After time is up, say: 'Keep your prediction — do NOT change it yet. We will come back to it.' Cold-call exactly 3 learners: one who chose A, one who chose B, one who chose C. Ask each: 'Tell me the reasoning behind your choice.' [WAIT TIME: 12 seconds after each response before moving on.] Do not confirm or deny any answer. Say: 'All three of you have given reasons that connect to real mathematics. Let us gather evidence to see who is closest to the truth.'	Anchored Prediction with Justification — learners commit to a written answer before instruction, activating prior knowledge of $\pi \approx 3.14159\dots$ and BODMAS from Lessons 5–6. Requiring a written sentence forces explicit reasoning rather than guessing, and preserves the prediction for later revision, making conceptual change visible.	Teacher scans written justifications during the 90-second window, looking for: (1) learners who correctly identify that $28\pi > 88$ or $28\pi < 88$ (showing they can order irrational and rational values); (2) learners who attempt any subtraction or estimation. Learners who write 'error = 0' reveal a misconception that rounding makes the values equal — flag these learners for targeted questioning during the Explain Phase.
Observe Phase  VIDEO: Kutambua na kupanga kwa thamani halisi - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES	Learners work in pairs to complete an Evidence Table of eight absolute value expressions, recording both their working and the final non-negative answer. The table includes: $ 7 $, $ -7 $, $ -3/4 $, $ 0 $, $ -28 $, $ \pi - 3 $ (use $\pi \approx 3.14159$), $ \sqrt{2} $ (use $\sqrt{2} \approx 1.41421$), and $ 88 - 28\pi $ (use $\pi \approx 3.14159$). For each expression, learners write: (i) Is	Distribute the printed or board-displayed Evidence Table. Say: 'Work with your partner. For every expression, first decide: is the number inside the absolute value sign positive, negative, or zero? That decision tells you the rule to use. You have 8 minutes.' [Circulate, observe pair work. Do not give answers. When a pair is	Structured Evidence Table with Pattern Noticing — the sequenced table moves from familiar integers to fractions, zero, and finally irrational expressions, allowing learners to induce the pattern ' $ x $ is always non-negative' through repeated application rather than being told the rule. The final entry $ 88 - 28\pi $ is the phenomenon re-	Teacher looks for: (1) correct sign analysis in column (i) — particularly for $ \pi - 3 $, where learners must recognise $\pi - 3 > 0$; (2) the common error of writing $ -7 = -7$; (3) whether pairs correctly classify $ 88 - 28\pi $ as irrational (since 88 is rational, 28π is irrational, their difference is irrational, and $ \text{irrational} =$

for similar videos EXERCISE: Number opposites Source: Khan Academy (English - US curriculum) Search ARES for similar readings	the number inside the $$ positive, negative, or zero? (ii) What is x? After completing the table, pairs compare their value for $88 - 28\pi$ with the class and discuss: 'Is this error rational or irrational? How do you know?' Learners record observations in the 'Observed' column of their summary table.	stuck on $\pi - 3$, prompt: 'Use your calculator or the value we have been using all term. Is 3.14159 minus 3 a positive or negative number?'] After 8 minutes, bring the class together. Write a blank version of the table on the board. Cold-call 5 different pairs — one per row — to give their answers, saying: 'Pair three — what did you get for $\pi - 3$ and how did you decide?' [WAIT TIME: 10 seconds per pair.] For the final entry $88 - 28\pi$, ask the whole class: 'Thumbs up if you got approximately 0.035. Thumbs sideways if you got something different.' Address discrepancies immediately by working through $28 \times 3.14159 = 87.96452\dots$ on the board, then $88 - 87.96452 = 0.03548\dots$ so $88 - 28\pi \approx 0.035$ cm.	entry point that makes the abstract concept immediately meaningful.	irrational). Any pair writing $88 - 28\pi = 0$ or a 'nice' fraction needs immediate intervention.
Explain Phase VIDEO: Placing absolute values on the number line Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Number opposites challenge Source: Khan Academy (English - US curriculum) Search ARES for similar readings	The teacher formally introduces the piecewise definition of absolute value: $x = x$ if $x \geq 0$; $x = -x$ if $x < 0$. Learners copy this into their notes and immediately apply it to verify three entries from their Evidence Table. They then work a set of four contextualised 'error' problems in their books: (1) A maize trader in Eldoret Market weighs a bag and reads 49.7 kg; the true mass is 50 kg. What is the error $50 - 49.7$? (2) A civil engineer in Thika specifies a steel rod of length $\sqrt{5}$ m ≈ 2.236 m; the manufactured rod is 2.24 m. What is $2.24 - \sqrt{5}$? (3) A pharmacist in Kisumu prepares a 0.5 mg tablet; the actual dose is 0.487 mg. What is $0.487 - 0.5$? (4) The daughter's string error revisited: $88 - 28\pi$ in exact symbolic form and as a decimal to 3 significant figures. For problem (4), learners write a	Write the piecewise definition on the board in coloured chalk — one colour for $x \geq 0$, another for $x < 0$. Say: 'This definition tells us: absolute value is just distance from zero — and distance is never negative. Let us check it against our Evidence Table. Pick any two entries and verify the definition gives the same answer you found.' [Allow 2 minutes.] Then say: 'Now I want to show you why this matters outside school. Every Kenyan professional who works with measurements uses this idea, even if they do not call it absolute value.' Work through the Eldoret maize problem aloud as a model, narrating every step. Then say: 'Problems 2, 3, and 4 are yours. You have 6 minutes. For problem 4, I want a symbolic exact answer AND a decimal answer. Bonus challenge: is the exact error for	Contextualised Application with Symbolic–Decimal Bridge — by requiring BOTH an exact symbolic form and a decimal approximation for each error, learners practise the core tension of the entire unit: precision vs. practicality. The Kenyan professional contexts (Eldoret trader, Thika engineer, Kisumu pharmacist) show that absolute value is not an abstract exercise but a real measurement tool, directly connecting to the driving question about how professionals 'work accurately with numbers that go on forever.'	Teacher checks that: (1) learners correctly apply the piecewise rule — particularly for $2.24 - \sqrt{5}$ where $2.24 < \sqrt{5}$ so the answer requires negating ($\sqrt{5} - 2.24$); (2) learners can state in writing that $88 - 28\pi = 28\pi - 88$ exactly because $28\pi \approx 87.965 > 88$ is FALSE — teacher must watch for this: $28 \times 3.14159 \approx 87.965$, so $88 > 28\pi$, meaning $88 - 28\pi = 88 - 28\pi \approx 0.035$; (3) learners connect the irrational nature of 28π to the irrational nature of the error. Exit ticket: learner writes one sentence — 'The string error is [rational/irrational] because...'

	sentence explaining whether the error is rational or irrational and why.	problem 2 rational or irrational?' [Circulate; targeted questioning for struggling learners: 'Is $\sqrt{5}$ bigger or smaller than 2.24? So is $2.24 - \sqrt{5}$ positive or negative? What does the definition say to do then?'] After 6 minutes, cold-call 4 learners — one per problem — asking them to read both their answer AND their reasoning aloud. [WAIT TIME: 12 seconds each.] Confirm: $88 - 28\pi = 28\pi - 88$ in exact form (since $28\pi > 88$), ≈ 0.0354 cm. Emphasise: 'This tiny gap is itself an irrational number — it cannot be written as a fraction. The daughter's string problem goes all the way down.'		
Driving Question Board (DQB) Creation  VIDEO: Comparing absolute values on the number line Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Vinyume vya namba mafumbo Source: Khan Academy (Kiswahili)  Search ARES for similar readings	Learners revisit the class Driving Question Board displayed at the front. In pairs, they write one updated sticky note (or card) responding to the prompt: 'We now know that the daughter's string error is $88 - 28\pi \approx 0.035$ cm — an irrational number. What does this add to our answer to the driving question: WHY can some numbers never be written down completely, and HOW do professionals work accurately with them?' Pairs also identify one question that Lesson 7 has raised for them that they think will be answered in Lesson 8 (the final lesson). These new questions are added to the 'Still Wondering' column of the DQB.	Point to the DQB and say: 'Look at what we added in Lessons 1 through 6. Today we added something new — we can now MEASURE the error. Take 3 minutes with your partner and write: first, one thing that Lesson 7 has changed or sharpened in your answer to the driving question; second, one question you still have.' Distribute cards or sticky notes. [Allow 3 minutes.] Invite 3 pairs to share their updated understanding — cold-call, [WAIT TIME: 10 seconds each]. Then ask: 'What question does the class most want answered in our final lesson?' Take a quick show of hands for the top two questions and physically place them in the centre of the DQB 'Still Wondering' column. Say: 'We have one lesson left. These two questions are our target for Lesson 8.'	DQB Progressive Revision — the Driving Question Board functions as a visible record of the class's growing explanatory model. Requiring learners to articulate what has CHANGED (not just what was learned) forces metacognitive comparison across lessons. The 'Still Wondering' column maintains productive intellectual tension and gives Lesson 8 a learner-generated purpose.	Teacher reads the sticky notes for evidence that learners connect absolute value to the phenomenon (not just list the definition). Strong responses will say something like: 'The error is irrational, so no matter how precisely the daughter cuts the string, there is always a gap — and that gap is itself a number that goes on forever.' Weak responses will repeat the definition without connecting to the phenomenon. Teacher notes pairs producing weak responses for follow-up in Lesson 8.
Model Building Phase  VIDEO:	Learners individually update their cumulative Real Numbers Model — a structured diagram they have been building since Lesson 1.	Say: 'Open your Real Numbers Model. We are adding the seventh layer — Absolute Value and Error. Follow these four steps.' Display	Cumulative Model Annotation — the running model, built across all eight lessons, externalises and preserves the learner's growing	Teacher checks models for three specific features: (1) correct placement of 28π to the LEFT of 88 on the number line ($28\pi \approx$

Compare rational numbers using a number line Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Complex numbers from absolute value & angle Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Today's addition is a new annotation layer titled 'Absolute Value & Error.' On their model, learners: (1) draw a number line segment showing 28π and 88 as two distinct points, labelling the gap as $88 - 28\pi = 28\pi - 88 \approx 0.035$ cm (since $28\pi > 88$ is FALSE — they mark 88 slightly to the right of 28π); (2) write the exact symbolic form of the error and classify it (irrational); (3) add an 'Absolute Value Rule' box: $x = x$ if $x \geq 0$; $x = -x$ if $x < 0$; (4) write one sentence connecting absolute value to the driving question. Learners who finish early add a second example: the error $2.24 - \sqrt{5}$ on a separate mini number line.	the four steps on the board. [Allow 5 minutes of individual model-building work.] Circulate and check that learners are placing 28π and 88 in the correct order on the number line — $28\pi \approx 87.965$ is to the LEFT of 88, so the gap $88 - 28\pi \approx 0.035$ is positive, and $88 - 28\pi = 88 - 28\pi$. Correct any learner who reverses this. Say to the class: 'Quick check — who can tell me: is 28π to the left or right of 88 on the number line? Hands down — I will cold-call.' [Cold-call 2 learners, WAIT TIME 10 seconds each.] Confirm: '$28\pi \approx 87.965$, so it is to the LEFT of 88. The string is 88 cm — a tiny bit LONGER than the exact circumference. The daughter will have a tiny overlap, not a gap.' Conclude: 'Your model now shows the complete journey: from classifying numbers in Lesson 1, all the way to measuring the exact error with absolute value today. One lesson remains to bring it all together.'	understanding as a single coherent artefact. Adding a geometrically accurate number line segment with both 28π and 88 marked forces learners to reconcile their arithmetic (which is larger?) with their spatial intuition, consolidating both the absolute value definition and the phenomenon's resolution. The model will be completed and presented in Lesson 8.	$87.965 < 88$); (2) the gap labelled as both $88 - 28\pi$ (exact, since positive) and ≈ 0.035 cm; (3) the classification of the error as irrational. A model that correctly shows all three features, with the connecting sentence linking to the driving question, meets expectations. A model missing the classification or reversing the number order approaches expectations and requires teacher follow-up.
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D. TEACHER REFLECTION

1. When learners evaluated $|88 - 28\pi|$, did they correctly determine that $88 > 28\pi$ (i.e. $88 > 87.965\dots$) and therefore apply the rule $|x| = x$ — or did a significant number reverse the order? If reversal was common, what does this reveal about their fluency with comparing irrational and rational values, and how will I address it in Lesson 8?
2. Were learners able to articulate — in writing — WHY the string error $|88 - 28\pi|$ is irrational? Did they connect this to the operations rules from Lessons 5 and 6 (rational $\pm$ irrational = irrational), or did they treat the irrational classification as a separate, disconnected fact?
3. How effectively did the Kenyan professional contexts (Eldoret maize trader, Thika engineer, Kisumu pharmacist) help learners see absolute value as a real tool rather than an abstract rule? Which context generated the most engagement, and should I lead with that context in Lesson 8's review?
4. Looking at the updated DQB, do the 'Still Wondering' questions learners generated point toward the content planned for Lesson 8 (synthesis, representation, and the full real number system), or have learners surfaced unexpected misconceptions that need to be addressed before synthesis?
5. Are learners' cumulative Real Numbers Models becoming coherent narratives that connect all seven lessons, or are they fragmented collections of rules? What structured support (sentence starters, a model checklist, a peer-review protocol) should I provide at the start of Lesson 8 to help all learners consolidate their models into a complete, driving-question-responsive explanation?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed that evaluating $ 88 - 28\pi $ using the piecewise definition of absolute value gives approximately 0.035 cm — a small but non-zero gap — and that this error is itself an irrational number. They also observed that absolute value always produces a non-negative result for any real number, whether integer, fraction, or irrational expression such as $ \pi - 3 $ or $ \sqrt{2} $.
What did I learn?	Absolute value $ x $ measures the distance of a real number x from zero on the number line and is defined as: $ x = x$ when $x \geq 0$, and $ x = -x$ when $x < 0$. The absolute difference $ a - b $ measures the error between any two real numbers. The daughter's string error is $ 88 - 28\pi = 88 - 28\pi \approx 0.035$ cm — rational minus irrational, which is irrational — meaning the error can never be expressed as an exact fraction either.
How does this explain the phenomenon?	The Ugali Pot Problem now has a precise, measurable answer: the daughter's 88 cm string is approximately 0.035 cm longer than the exact circumference 28π cm. This tiny overlap is itself an irrational number — it goes on forever without repeating. No matter how many decimal places of π she uses, the string error will always be a non-terminating, non-repeating number. Absolute value gives mathematicians and Kenyan professionals (engineers, traders, pharmacists) a rigorous, always-positive way to state 'how wrong' any rational approximation of an irrational measurement really is.

LESSON 8 (80 minutes): Absolute Value: Distance from Zero — and the Final Explanation of the Real Number System

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners consolidate the entire real number sub-strand by connecting absolute value to the number line, applying it in Kenyan real-life contexts, and constructing a final, comprehensive explanation of why some numbers cannot be written down completely — answering the driving question in full.
Knowledge	Learners will know: (1) the formal definition of absolute value $ x $ as the distance of x from zero on the number line; (2) that absolute value is always non-negative; (3) that absolute value can be applied to both rational and irrational real numbers; (4) how to evaluate absolute value expressions including those involving temperature changes, profit/loss in shillings, and circumference approximations; (5) the complete classification of real numbers (natural, whole, integer, rational, irrational) and where absolute value fits within that system.
Skills	Learners will be able to: (a) compute $ x $ for integers, fractions, decimals and irrational numbers; (b) represent absolute value results on a number line; (c) solve simple absolute value expressions in context; (d) compare the absolute values of rational and irrational numbers; (e) construct a final revised model of the real number system incorporating all sub-strand concepts; (f) use mathematical reasoning to articulate a complete, evidence-based answer to the driving question.
Attitudes	Learners will appreciate: (a) the power of precise mathematical language to describe everyday phenomena such as temperature swings in Kericho or profit/loss at Gikomba Market; (b) that mathematics is a tool for making accurate decisions even when exact values are impossible to write; (c) the collaborative value of revisiting and refining ideas over the course of a unit; (d) intellectual honesty in acknowledging approximation as a professional and mathematical necessity.
Key Inquiry Question	What are real numbers? How do we use real numbers in day-to-day situations?
Purpose in Storyline	This is the final lesson of the sub-strand. The Kericho temperature supporting phenomenon is revisited — learners determine which day had the largest temperature swing away from zero, discovering absolute value as 'distance from zero on the number line.' They then return to the Ugali Pot anchoring phenomenon to deliver their Final Explanation: why the circumference 28π cannot be written exactly, how absolute value applies to both rational and irrational numbers equally, and how professionals — engineers, nurses, market traders, meteorologists — work accurately with numbers that go on forever. The DQB is completed, and individual models from Lesson 1 are compared with the final class model.
Safety Notes	No hazardous materials. Ensure learners handle rulers and pencils carefully during number line construction activities. If using printed thermometer cards, ensure no sharp edges on laminated materials.

B. LESSON OVERVIEW

Lesson 8 is the capstone lesson for Sub-Strand 1.1: Real Numbers. It opens by revisiting the Kericho temperature data supporting phenomenon (daily high/low temperatures recorded over a week) to motivate absolute value as 'distance from zero.' Learners predict which day had the biggest temperature swing, then compute absolute values formally and represent results on number lines. Through a structured Explain phase, learners discover that absolute value applies equally to rational numbers (Ksh 60.50 price change) and irrational numbers (π -based circumference error), deepening their understanding that absolute value is a property of ALL real numbers. The DQB is formally closed with evidence-backed answers, and learners compare their Lesson 1 model of the number system with the class's final, comprehensive model. The lesson concludes with a whole-class Final Explanation of the driving question, synthesising all eight lessons of evidence into one coherent answer. NGSS Science and Engineering Practice 6 (Constructing Explanations) and Practice 4 (Analysing and Interpreting Data) are foregrounded throughout.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Meaning of absolute value Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Ordering negative number expressions Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners are shown a table of Kericho daily temperatures for one week (Monday–Sunday), with both morning low and afternoon high temperatures recorded in °C. Some values are negative (cold mornings: e.g., -3.5°C on Wednesday) and some positive (warm afternoons: e.g., 28°C on Saturday). Learners work individually for 3 minutes, then discuss in pairs for 2 minutes, to answer: 'Which day had the BIGGEST swing away from zero degrees? How would you measure that swing mathematically?' They record predictions and reasoning in their exercise books. Learners also revisit the Ugali Pot phenomenon by answering: 'The daughter's calculator showed 87.96459430... and she cut 88 cm. How far off was her cut — and does it matter whether that error involves a rational or irrational number?'	1. Display the Kericho temperature table on the board (or distribute printed cards). Read aloud: 'A farmer in Kericho recorded these temperatures. Monday morning: 2°C, Monday afternoon: 19°C. Wednesday morning: -3.5°C, Wednesday afternoon: 24°C. Saturday morning: 1°C, Saturday afternoon: 28°C.' Ask: 'Before we calculate anything — which day do you think had the biggest swing AWAY from zero? Write your prediction and your reason. You have 3 minutes — go.' Set a visible timer. 2. After 3 minutes, say: 'Turn to your partner. Share your prediction. Do you agree? You have 2 minutes.' 3. Cold-call 3 pairs. Ask each: 'What did you predict, and what was your reasoning?' Wait 10 seconds after each response before commenting. 4. Do NOT confirm or deny. Say: 'Hold your predictions — we are going to find a mathematical tool that answers this precisely.' 5. Then display the Ugali Pot follow-up prompt: 'The daughter cut 88 cm but the exact value is 87.96459430... cm. In your book, estimate: how far off is 88 from the true value? Write ONE number.' Wait 15 seconds. Cold-call 2 learners. Accept all answers without evaluation. 6. Say: 'Today we build the mathematical tool that measures EXACTLY how far a number is from zero — and then we use it to close every question on our Driving Question Board.'	Activate-and-Connect Prediction: Learners draw on their existing intuition about 'distance' and 'swing' before formal vocabulary is introduced. By connecting both the Kericho temperature context and the Ugali Pot error context, learners immediately see that the new concept (absolute value) has dual relevance — it applies to integers/decimals (rational) and to irrational approximation errors alike. This primes cognitive readiness for the formal definition.	Observable indicator: Each learner has written a numerical prediction and a written reason in their book within 3 minutes. Teacher circulates and checks that learners are attempting to compare magnitudes (not just reading the largest positive number). Listen for misconceptions: 'The biggest swing is Saturday because 28 is the biggest number' — this reveals learners are not yet thinking about distance from zero for negative values. Flag these learners for targeted questioning in Phase 2.
Observe Phase	Learners engage in a structured data analysis activity using the	1. Distribute A4 number line sheets (pre-printed with a number	Representational Bridging (Number Line to Symbol):	Observable indicators: (1) All number lines show correct

 VIDEO: Kutambua na kupanga kwa thamani halisi - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  EXERCISE: Number opposites Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Kericho temperature dataset. In groups of 4, they: (a) plot all 14 temperature values (7 mornings + 7 afternoons) on a shared number line drawn on A4 paper, marking each value with a dot and labelling it; (b) draw arrows from zero to each value and measure the arrow length (distance from zero), recording results in a table; (c) notice that -3.5 and $+3.5$ are the SAME distance from zero; (d) use their calculators to compute $28 - 0 = 28$ and $-3.5 - 0 = 3.5$ and compare. Then individually, learners complete an Observation Recording Sheet with three columns: 'Temperature value', 'Distance from zero (arrow length)', 'Which is bigger: the morning low distance or the afternoon high distance?' Finally, learners observe a second dataset: Gikomba Market daily profit/loss for Mama Wanjiku's unga stall over 5 days (e.g., +Ksh 350, -Ksh 120, +Ksh 580, -Ksh 75, +Ksh 210). Learners compute the distance from zero (Ksh 0) for each value and identify the day of biggest financial swing.	line from -30 to $+30$ in steps of 5). Say: 'In your groups, plot every temperature value from the table onto this number line. Take 5 minutes. Every member must plot at least two values.' Circulate — check that negative values are placed correctly to the left of zero. 2. After plotting, say: 'Now draw an arrow from zero to each dot. The LENGTH of that arrow is the DISTANCE from zero. Measure it by counting steps. Record it in your table.' Allow 4 minutes. 3. Ask the whole class: 'What do you notice about -3.5 and $+3.5$ on your number line?' Wait 10 seconds. Cold-call 2 groups. Target the response: 'They are the same distance from zero.' If not offered, prompt: 'Look at the arrows — are they the same length or different?' 4. Write on the board: 'Distance from zero = x. We call this the ABSOLUTE VALUE of x.' Say: 'The vertical bars mean: ignore the sign — only keep the size.' 5. Demonstrate: $-3.5 = 3.5$, $28 = 28$, $0 = 0$. Ask learners to verify with their arrows. 6. Introduce the Gikomba profit/loss dataset. Say: 'Mama Wanjiku's stall made these profits and losses. In your book, find the absolute value of each day's result, then answer: on which day was the financial swing BIGGEST?' Allow 4 minutes. Cold-call 3 learners for answers and reasoning. Wait 12 seconds after each. 7. Conclude Observe phase by asking: 'Can you find $\pi - 3.14$? Try it on your calculator.' Allow 2 minutes. Say: 'We will use this exact idea to measure the Ugali Pot cutting error very soon.'	Learners first build the concept physically — drawing arrows on a number line — before encountering the algebraic symbol x. The arrow length IS the absolute value. This grounding in physical representation prevents the common misconception that absolute value 'just removes the minus sign' without understanding why. The Gikomba profit/loss context extends the concept from temperature (a natural 'distance from zero' context) to financial mathematics, reinforcing transferability across real-life Kenyan contexts.	placement of negative values to the left of zero. (2) Learners' tables show equal distances for equal-magnitude positive and negative values. (3) When computing $\pi - 3.14$, learners correctly get approximately 0.00159... — indicating they can apply absolute value to irrational contexts. Teacher listens for the misconception 'absolute value is always positive' stated without understanding — probe with: 'What is 0? Does your rule still work?' Exit check: cold-call one learner to state the formal definition in their own words.
Explain Phase  VIDEO:	Learners work through a structured sensemaking sequence	1. PART A: Write the formal definition on the board in full. Say:	Evidence-to-Explanation Progression (NGSS SEP 6 —	Observable indicators: (1) Learners correctly write both cases

[Placing absolute values on the number line](#)

Source: Khan Academy (English - US curriculum)
[Search ARES for similar videos](#)

 EXERCISE:

[Number opposites challenge](#)

Source: Khan Academy (English - US curriculum)
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in three parts. PART A — Formal Definition and Cases: Learners copy and complete a structured note: 'The absolute value of a real number x , written $|x|$, is its distance from zero on the number line. Formally: $|x| = x$ if $x \geq 0$, and $|x| = -x$ if $x < 0$.' They verify this definition with six examples: $|7|$, $|-12|$, $|0|$, $|-\sqrt{2}|$, $|3/4|$, $|-Ksh\ 120|$. PART B — Application Problems in Kenyan Context: Learners solve five contextualised problems: (1) The temperature in Kericho fell from 24°C to -3.5°C overnight. What is $|24 - (-3.5)| = |27.5| = 27.5^{\circ}\text{C}$? (2) A boda boda rider in Kisumu owes Ksh 450 on his loan. His account balance is -450 . What is $|-450|$? (3) The exact circumference of the sufuria is 28π cm. A learner approximates it as 88 cm. The error is $|28\pi - 88| = |87.9646... - 88| = |-0.0354...| \approx 0.035$ cm. Is this error rational or irrational? (4) A nurse in Mombasa measures a patient's temperature as 37.8°C . Normal is 37°C . What is the deviation $|37.8 - 37| = 0.8^{\circ}\text{C}$? (5) On a number line, plot $|\sqrt{3}|$ and $|-\sqrt{3}|$ and explain why they give the same point. PART C — DQB Question Resolution: The class returns to the Driving Question Board. Each sticky note question is read aloud. Learners vote (thumbs up/sideways/down) on whether it has now been answered. For each answered question, a volunteer writes a one-sentence evidence-based answer on a green card placed next to the sticky note.

'Copy this definition exactly — it is the formal mathematical language. Then verify it works for each of my six examples. You have 5 minutes.' After 5 minutes, cold-call 2 learners to read their answers for $|-\sqrt{2}|$ and $|3/4|$. Wait 10 seconds each. Correct any errors immediately with: 'What is the distance from zero to $-\sqrt{2}$ on the number line? Draw the arrow if you are unsure.' 2. PART B: Say: 'Now we use absolute value in real Kenyan situations. Work individually. You have 10 minutes for all five problems.' Circulate — target Problem 3 (the irrational error) to check learners compute $|28\pi - 88|$ correctly. If learners struggle, prompt: 'What does 28π equal on your calculator? Now subtract 88. Now take the absolute value.' 3. After 10 minutes, cold-call one learner per problem. For Problem 3, ask: 'Is $28\pi - 88$ rational or irrational? Is $|28\pi - 88|$ rational or irrational? What does that tell us about absolute value and irrational numbers?' Wait 15 seconds. Accept and validate: 'Correct — absolute value does NOT change whether a number is rational or irrational. It only changes sign, never the number type.' 4. PART C — DQB: Read each sticky note question aloud. For each, say: 'Who thinks we have answered this? Thumbs up if yes, sideways if partially, down if not yet.' For unanswered questions, say: 'Is this something we can answer now, or is it beyond Grade 10 Real Numbers?' Acknowledge genuinely open questions (e.g., 'Are there more irrational numbers than rational?') as examples of mathematical curiosity that leads to further study.

Constructing Explanations): Learners move systematically from observed evidence (arrow lengths on number lines, calculator outputs) through structured application (five contextualised problems) to formal conceptual closure (DQB resolution). Problem 3 deliberately bridges the absolute value concept back to the anchoring phenomenon, ensuring the new tool (absolute value) is understood as a component of the broader real number story — not an isolated technique. The thumbs voting on DQB questions makes each learner's evolving understanding visible and accountable.

of the formal definition ($x \geq 0$ and $x < 0$). (2) Problem 3 responses show correct calculator use: $28 \times \pi \approx 87.9646$, subtracted from 88, absolute value taken. (3) DQB vote is not unanimous — some learners' 'sideways' votes are valuable data about residual confusion. Teacher notes which questions remain 'sideways' for targeted wrap-up. (4) The one-sentence DQB answer about absolute value and rational/irrational numbers includes a specific example (not just a statement).

		5. Close Part C by asking: 'Does absolute value change whether a number is rational or irrational? Someone give me a one-sentence answer with an example.' Cold-call 1 learner. Wait 12 seconds.		
Driving Question Board (DQB) Creation  VIDEO: Comparing absolute values on the number line Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Vinyume vya namba mafumbo Source: Khan Academy (Kiswahili)  Search ARES for similar readings	This is the formal Final Explanation phase. Learners participate in a whole-class structured discussion to construct a shared, comprehensive answer to the driving question: 'Why can we never write some everyday numbers down completely — and how do mathematicians and Kenyan professionals work accurately with numbers that go on forever?' The teacher facilitates a 'Evidence Gallery Walk' around the classroom, where posters from previous lessons (number classifications, rational/irrational examples, decimal expansions, operations, reciprocals, the number line) are displayed. Each learner carries a 'Final Explanation Card' — a half-page template with three sentence starters: (1) 'Some numbers cannot be written down completely because...'; (2) 'The real number system is organised as...'; (3) 'Kenyan professionals handle this by...'. Learners complete their cards individually (5 minutes), then share in groups of 4 (3 minutes). One group representative presents to the class. The class then co-constructs a single 'Final Class Explanation' written on a large poster, which is displayed alongside the now-complete DQB. Learners also compare their individual Lesson 1 model of the number system (drawn from memory in Lesson 1) with the class's final model, identifying at	1. Say: 'We have spent eight lessons on one big question. Today you answer it — completely, with evidence, like a mathematician.' Distribute Final Explanation Cards. Say: 'Complete all three sentence starters. Use specific numbers and examples from Kenya. You have 5 minutes — work in silence.' Set timer. Circulate and read over shoulders — note strong responses for cold-calling. 2. After 5 minutes: 'Share your card in your group of 4. Each person reads aloud. After all four have read, agree on the BEST idea from each sentence starter. You have 3 minutes.' 3. Cold-call one group representative per sentence starter (3 cold-calls total). After each, ask the class: 'Does anyone want to add one word or one example to make this stronger?' Wait 10 seconds. Accept 1–2 additions maximum per starter. 4. With learner contributions, write the Final Class Explanation on the poster in real time. Read it aloud when complete. 5. Display learners' Lesson 1 models (collected and stored by teacher). Say: 'Look at what you drew in Lesson 1. In your book, write: (a) One thing your Lesson 1 model got right. (b) Two things you would change or add now.' Allow 3 minutes. Cold-call 2 learners: 'What would you add to your Lesson 1 model?' Wait 12 seconds each. 6. Point to the DQB. Say: 'Every question we	Cumulative Model Comparison (NGSS SEP 2 — Developing and Using Models): By physically comparing their Lesson 1 model with the final class model, learners make their own intellectual growth visible and concrete. The Final Explanation Card's three sentence starters scaffold a complete scientific explanation: mechanism ('because...'), system description ('organised as...'), and application ('professionals handle this by...'). The Gallery Walk anchors the explanation in accumulated evidence rather than memory alone, honouring the full eight-lesson storyline.	Observable indicators: (1) All three sentence starters are completed with specific mathematical content (not vague generalities). (2) Learners identify irrational numbers (not just 'hard numbers') as the class that cannot be written completely. (3) Model comparisons name specific additions — e.g., 'I would add irrational numbers as a separate branch,' 'I would show that integers are a subset of rationals,' 'I would label where $\sqrt{2}$ sits.' (4) At least one learner references the Ugali Pot phenomenon and at least one references the Kericho temperature in their Final Explanation Card, unprompted.

	least two specific improvements to their original drawing.	posted over eight lessons — let us read them one final time.' Read each aloud. For any still unanswered, say: 'This is a beautiful open question. It lives beyond Grade 10. Write it in your book as a question to explore in Grade 11 or university.' 7. Close with: 'The daughter at Gikomba — she could not write the circumference exactly. But now you can explain WHY. And you can tell her exactly how far off her string was, using absolute value. That is what mathematicians do.'		
Model Building Phase  VIDEO: Compare rational numbers using a number line Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Absolute value to find distance Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners individually draw their final, revised model of the real number system in their exercise books. The model must include: (a) all five number sets nested correctly (Natural $\subset$ Whole $\subset$ Integer $\subset$ Rational; Irrational $\cup$ Rational = Real); (b) at least two Kenyan examples in each set (e.g., Natural: 28 cm diameter of sufuria, 160 km Nairobi–Nakuru; Integer: -3.5°C Kericho morning rounded to -4; Rational: Ksh 60.50 per kg of unga, 34.5°C Kisumu temperature; Irrational: π, $\sqrt{2}$, 28π); (c) a number line from -5 to $+5$ with absolute value arrows shown for at least three values; (d) one sentence explaining where absolute value 'lives' in the system — 'Absolute value applies to ALL real numbers — it measures distance from zero regardless of whether the number is rational or irrational.' Learners then complete a 3-2-1 Exit Ticket: 3 things learned in this sub-strand, 2 real Kenyan situations where this matters, 1 question still puzzling them.	1. Say: 'This is your final model — make it your best work. Use the poster, the DQB, and your notes. You have 10 minutes. Every part of the checklist must appear.' Write the five checklist items on the board. Circulate and check: (a) Are nested sets correctly drawn? (b) Are Kenyan examples specific and correct? (c) Is the number line labelled with absolute value arrows? (d) Is the absolute value sentence present? 2. After 10 minutes, cold-call 2 learners to share their models under the document camera (or hold up for class to see). Ask each: 'Where did you place 28π on your model? Why?' Wait 10 seconds. 3. Say: 'Now complete your 3-2-1 Exit Ticket. Be specific — I should be able to read this and know exactly what YOU understand.' Allow 5 minutes. Collect all exit tickets. 4. Final close: Return to the phenomenon one last time. Say: 'The daughter at Gikomba cut 88 cm of string. The exact value is 28π cm. She will always have a tiny gap. Now you can say: the gap is $88 - 28\pi \approx 0.035$ cm — a RATIONAL number that	Synthesis Model Construction with Checklist Accountability: The five-part checklist ensures the final model is genuinely comprehensive rather than a rushed sketch. The checklist items map directly to the sub-strand's specific learning outcomes (classification, representation, operations, real-life application, absolute value), making the model a visual SLO-completion artefact. The 3-2-1 Exit Ticket separates content consolidation (3 things learned), application (2 Kenyan situations), and metacognitive monitoring (1 remaining question) — giving the teacher diagnostic data for the start of the next sub-strand.	Observable indicators: (1) Final models show correct nesting (Natural inside Whole inside Integer inside Rational, with Irrational as a separate non-overlapping branch, both inside Real). A common error — placing Irrational inside Rational — is immediately visible. (2) Number lines show arrows drawn FROM zero TO the plotted value, with the length labelled as x. (3) Exit Ticket '2 Kenyan situations' responses should name specific professional contexts (not generic 'in maths class'). (4) Exit Ticket '1 question' responses are collected and reviewed by teacher to inform the opening of Sub-Strand 1.2 (Indices and Logarithms) in the next lesson cycle.

		approximates an IRRATIONAL error. And that is exactly how engineers at the Kenya Railways, nurses at Kenyatta National Hospital, and farmers in the Rift Valley work every day — they use the best rational approximation available, and they know precisely how close it is. That is the answer to our driving question.' 5. Formally close the DQB: 'Our Driving Question Board is complete. Every question has been addressed. Well done.'		
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D. TEACHER REFLECTION

After this lesson, reflect on the following questions: (1) Did learners demonstrate a genuine shift from their Lesson 1 model to their final model — specifically, did they correctly separate Rational and Irrational as distinct, non-overlapping subsets of Real? Review collected Lesson 1 vs. final model pairs and count how many learners made this correction. (2) In the Final Explanation Cards, how many learners spontaneously referenced the Ugali Pot phenomenon without prompting? This is a strong indicator that the anchoring phenomenon successfully anchored learning across all 8 lessons. (3) Were learners able to compute $|28\pi - 88|$ correctly, or did many leave the absolute value as a negative number? If the latter, revisit the formal definition at the start of Sub-Strand 1.2. (4) Review the '1 question still puzzling me' entries from Exit Tickets: common themes (e.g., 'Are there different sizes of infinity in irrational numbers?') can be acknowledged in the next lesson as rich mathematical territory. (5) Consider whether the DQB process genuinely empowered learner voice — did learners post questions in early lessons that they themselves answered in later lessons? If most DQB questions were answered only by the teacher, redesign early DQB phases to be more learner-led in the next sub-strand cycle.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners plotted Kericho weekly temperature values (including negatives such as -3.5°C) on a number line, drew arrows from zero to each value, and measured arrow lengths. They also computed absolute values of Gikomba Market daily profit/loss figures and calculated $ 28\pi - 88 \approx 0.035$ cm as the Ugali Pot cutting error.
What did I learn?	Absolute value $ x $ is the distance of a real number x from zero on the number line. It is always non-negative. Formally: $ x = x$ if $x \geq 0$, and $ x = -x$ if $x < 0$. Absolute value applies equally to rational numbers (e.g., $ -Ksh\ 450 = 450$) and irrational numbers (e.g., $ \sqrt{2} = \sqrt{2}$, $ 28\pi - 88 \approx 0.035$). It does NOT change whether a number is rational or irrational. The complete real number system classifies all numbers as either rational or irrational, and absolute value is a tool that works across the entire system to measure magnitude without regard to direction.
How does this explain the phenomenon?	Some everyday numbers — such as the circumference of a sufuria with diameter 28 cm — cannot be written down completely because they are irrational: their decimal expansions neither terminate nor repeat. The real number system contains both rational numbers (expressible as fractions, with terminating or repeating decimals, like Ksh 60.50 or 34.5°C) and irrational numbers (like π and $\sqrt{2}$, which arise naturally from geometry and other mathematics). Mathematicians and Kenyan professionals — engineers, nurses, market traders, meteorologists — work accurately by using the best available rational approximation and by quantifying the approximation error using absolute value, so they always know exactly how close their working value is to the true value.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.